An Uncertainty-Aware Multi-Fidelity Surrogate Modeling Framework for Inverse Problems with Application to Guided-Wave-based Damage Detection

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Abstract

Structural health monitoring relies on inverse problems to infer damage characteristics from measured responses. These inverse problems are typically ill-posed, with multiple damage configurations producing similar sensor readings, making uncertainty quantification essential for reliable damage assessment. Bayesian inference provides a principled framework for such inverse problems but requires thousands of forward model evaluations, which becomes computationally prohibitive when high-fidelity numerical simulations are employed.

This thesis develops an uncertainty-aware multi-fidelity surrogate modeling framework for inverse damage/parameter characterization and shows its application in guided-wave-based structural health monitoring. The framework combines a computationally expensive two-dimensional finite element model (high-fidelity) with a fast one-dimensional zigzag beam theory model (low-fidelity) through an uncertainty-aware transfer learning approach. An autoencoder compresses the wave responses into a latent space, initially trained on abundant low-fidelity samples and subsequently fine-tuned on limited high-fidelity samples. An XGBoost regressor maps damage parameters to latent representations, following the same two-phase training strategy. Surrogate-induced uncertainty arising from limited high-fidelity data is explicitly modeled through a Gaussian distribution in the latent space and propagated to the response space via Monte Carlo sampling.

The multi-fidelity surrogate achieves an R^2 of 0.9693 on high-fidelity test data, compared to 0.8703 for a high-fidelity-only surrogate trained on the same training samples. Bayesian inverse inference using Differential Evolution-initialized Parallel Tempering MCMC successfully estimates notch location and depth, with sensitivity analysis confirming these as the dominant parameters. The framework captures inherent ambiguities in wave-based inverse problems through multimodal posterior distributions, providing uncertainty-aware damage characterization that acknowledges the limitations of both the measurement data and the surrogate approximation.

Keywords: *Damage Detection, Multi-fidelity Modeling, Surrogate Model, Bayesian Inference, Guided Waves, Uncertainty Quantification*

Chapter 1

Introduction and Motivation

1.1 Inverse Problems and Uncertainty in Structural Health Monitoring

Engineering systems inevitably develop damage during service. Aircraft fuselages and pipelines may develop cracks, ship hulls undergo corrosion, and bridges accumulate defects over long periods of loading and environmental exposure. The practical challenge is therefore not merely to detect the presence of damage, but to characterize it sufficiently early to support informed maintenance and prevent catastrophic failure. This challenge naturally gives rise to *inverse* problems, in which measured structural responses are used to infer the underlying state of the system.

Structural Health Monitoring (SHM) addresses this problem by estimating damage characteristics—such as location, size, or severity—from response data collected using instrumented sensors during operation. These inverse problems are typically ill-posed: different damage configurations may produce similar measured responses, and only limited information is available from a small number of sensors. As a result, inverse solutions are often non-unique and sensitive to noise and modeling approximations.

In contrast, the corresponding *forward* problem is comparatively well posed. Given a structural model, material properties, and specified damage parameters, the structural response can be predicted using physics-based models. Inverse damage detection, however, requires repeated evaluation of such forward models for different candidate damage configurations, which becomes computationally demanding when high-fidelity numerical simulations are employed.

Uncertainty is therefore intrinsic to inverse problems in SHM. In the context of this work, uncertainty arises primarily from limited availability of high-fidelity simulation data and the approximation error introduced by surrogate models trained under data-scarce conditions. If this uncertainty is not explicitly accounted for, inverse estimates may appear overly precise while being fundamentally unreliable.

Deterministic inverse approaches typically formulate damage detection as an optimization problem and return a single best-fit solution. While such methods can yield plausible estimates, they do not quantify uncertainty or reveal alternative damage scenarios consistent with the data. Bayesian inference offers a principled alternative by inferring a probability distribution over damage parameters. However, sampling-based Bayesian methods such as Markov Chain Monte Carlo (MCMC) require thousands of forward model evaluations, motivating the need for efficient surrogate-based forward models.

1.2 Computational and Physical Challenges in Guided-Wave Damage Detection

The challenges associated with damage detection in SHM depend strongly on the sensing modality and the underlying physics of the measured response. Low-frequency global response methods, such as vibration-based techniques, are computationally efficient but provide limited spatial resolution, reducing sensitivity to small or localized defects. High-frequency wave-based methods, by contrast, offer enhanced damage sensitivity but introduce significant modeling and inference challenges.

Wave-based SHM relies on the interaction of propagating waves with structural defects. The resulting time-domain responses are dispersive and strongly dependent on damage characteristics, material properties, and geometry. Accurately capturing these wave–damage interactions requires forward models with fine spatial and temporal resolution, leading to high computational cost.

Among wave-based approaches, guided waves in plate-like structures constitute a widely studied and practically relevant SHM technique. Guided waves can interrogate large regions using a limited number of actuator–sensor pairs, making them attractive for early-stage damage detection. At the same time, their sensitivity to damage complicates inverse inference, as different damage configurations may produce subtle and ambiguous variations in the measured waveforms.

From an inverse problem perspective, baseline-free guided-wave damage detection is particularly challenging. Without a pristine reference state, damage characterization relies entirely on learning changes in the waveform as a function of damage parameters. Different damage scenarios may generate similar responses, leading to non-unique inverse solutions and, in a probabilistic setting, multi-modal posterior distributions. Reliable inference therefore requires repeated evaluation of forward models, which is infeasible when high-fidelity simulations are used directly.

High-fidelity numerical models accurately resolve wave–damage interactions but are computationally expensive, and only a limited number of such simulations are typically available. Surrogate models trained under these data-scarce conditions inevitably introduce approximation errors. If surrogate-induced uncertainty is ignored, inverse inference may result in overconfident or biased damage estimates.

The choice of model fidelity thus plays a central role in inverse damage detection. High-fidelity models provide accuracy but are unsuitable for repeated use in Bayesian inference, while low-fidelity models reduce computational cost at the expense of systematic error. This trade-off motivates the development of uncertainty-aware multi-fidelity surrogate modeling strategies. Guided-wave–based SHM is adopted in this thesis as a representative and challenging testbed for developing and assessing such inverse inference frameworks, while the proposed methodology remains applicable to a broader class of SHM problems.

1.3 Multi-Fidelity Modeling for Surrogate-Based Inference

Surrogate models approximate the input–output relationship of computationally expensive numerical models, with the hope that the surrogate model runs much faster than the original expensive numerical model, while still retaining sufficient accuracy away from trained data points. In our case, instead of running a finite element simulation for every candidate set of damage parameters, a surrogate model provides a fast approximation of the corresponding structural response. Once trained, surrogate evaluations typically take milliseconds rather than minutes, which makes Bayesian inference computationally feasible in practice.

However, the use of surrogate models introduces additional challenges. A surrogate is only reliable within the region of the parameter space that is adequately represented in its training data. If important regions are sparsely sampled or entirely missing, surrogate predictions in those areas may be inaccurate. In inverse problems, this modeling error becomes critical. If surrogate uncertainty is ignored, the inferred damage parameters may appear much more certain than they truly are. Such overconfident posterior distributions are misleading and can result in poor decision-making, sometimes worse than having no uncertainty quantification at all.

Deterministic surrogates are particularly problematic; they provide single predicted responses with no accuracy indication. Probabilistic surrogates predict distributions over possible responses, which, when used with Bayesian inference, provide uncertainty.

Multi-fidelity modeling extends surrogate-based approaches by combining models of differing accuracy and computational cost. A limited number of high-fidelity simulations provide accurate reference solutions, while a larger set of low-fidelity simulations captures broader trends across the parameter space. When strong correlation exists between fidelity levels, multi-fidelity learning can achieve accuracy close to that of the high-fidelity model at substantially reduced computational cost.

In the context of inverse problems, multi-fidelity surrogate modeling enables uncertainty-aware Bayesian inference without prohibitive computational cost. The approach addresses the primary bottleneck of repeated forward model evaluations while preserving a principled treatment of uncertainty. As a result, the inferred posterior distributions reflect genuine uncertainty arising from data limitations and modeling error rather than artificial confidence introduced by overly simplistic approximations.

1.4 Research Gaps in Uncertainty-Aware Multi-Fidelity SHM

Multi-fidelity modeling and probabilistic learning have been investigated in the SHM and model calibration literature, including studies on multi-fidelity physics-informed learning for probabilistic damage diagnosis and multi-fidelity Bayesian calibration using MCMC. However, important gaps remain for inverse damage characterization problems driven by high-dimensional guided-wave time-series data.

First, many existing probabilistic multi-fidelity SHM approaches focus on diagnosis metrics or low-dimensional response quantities, whereas guided-wave damage identification requires learning and inverting full dispersive waveforms. This creates a need for surrogate models that are simultaneously (i) accurate for high-dimensional waveform outputs, (ii) fast enough to be embedded within sampling-based Bayesian inversion, and (iii) equipped with explicit representation of surrogate-induced uncertainty arising from limited high-fidelity data.

Second, while uncertainty is often acknowledged, fewer frameworks construct a likelihood for Bayesian inversion that explicitly incorporates surrogate prediction uncertainty in addition to measurement noise, leading to the risk of overconfident or biased posterior estimates when trained with sparse high-fidelity simulations. Moreover, the manner in which surrogate uncertainty propagates across fidelity levels—and how this impacts posterior multimodality and parameter identifiability in inverse damage detection—remains insufficiently characterized.

1.4.1 Problem Statement

There is a need for an uncertainty-aware multi-fidelity surrogate framework for guided-wave inverse damage characterization that (i) learns efficient surrogates for high-dimensional waveforms using a low-to-high fidelity hierarchy, and (ii) explicitly quantifies and propagates surrogate-induced uncertainty within a Bayesian inference pipeline to enable MCMC-based parameter estimation under limited high-fidelity data availability.

1.5 Research Objectives

The primary objective of this thesis is to develop an uncertainty-aware multi-fidelity surrogate modeling framework for inverse damage characterization under limited high-fidelity data availability. The specific objectives are:

- To develop a multi-fidelity surrogate modeling strategy that combines low- and high-fidelity physics-based models for efficient forward predictions.
- To incorporate explicit representation of surrogate approximation uncertainty arising from sparse high-fidelity training data.
- To enable computationally tractable Bayesian inverse inference by embedding the proposed surrogate model within a sampling-based framework.
- To examine the influence of surrogate accuracy and uncertainty on parameter identifiability and posterior distributions in inverse parameter estimation.
- To demonstrate the proposed methodology on a guided-wave damage detection problem involving a thin, long panel modeled under plane strain using a single actuator–sensor pitch–catch configuration.

1.6 Scope and Thesis Organization

The proposed framework is demonstrated using guided-wave–based damage detection in thin, long panel-like structures modeled under a plane strain assumption. A single actuator–sensor pair in a pitch–catch configuration is considered, where ultrasonic guided waves propagate through the structure, interact with localized damage, and are recorded at the sensor. The inverse problem addressed in this work consists of inferring damage parameters from the measured wave response within a probabilistic inference framework.

This application serves as a representative and challenging testbed for inverse inference under data scarcity. High-fidelity finite element models accurately capture wave–damage interactions but are computationally impractical for repeated use in Bayesian inference. Low-fidelity models reduce computational cost but introduce systematic modeling errors. When surrogate models are trained using a limited number of high-fidelity simulations, approximation errors are inevitable and must be explicitly accounted for. Accordingly, this thesis focuses on incorporating surrogate-induced uncertainty within a multi-fidelity learning framework to enable reliable Bayesian inverse inference.

Chapter 2 reviews background material on inverse problems in SHM, guided-wave propagation modeling, surrogate modeling, multi-fidelity learning, and Bayesian inference. Chapter 3 presents the uncertainty-aware multi-fidelity surrogate framework, including fidelity definitions, surrogate construction, and stochastic latent-space modeling. Chapter 4 describes Bayesian inverse inference using the proposed surrogate, including likelihood formulation, MCMC implementation, and posterior diagnostics. Chapter 5 presents the guided-wave application and associated numerical studies. Chapter 6 discusses results, limitations, and insights. Chapter 7 concludes the thesis and outlines future research directions.

The central contribution of this thesis is demonstrating that uncertainty-aware multi-fidelity surrogate modeling enables practical Bayesian inverse inference for damage detection when high-fidelity forward models are computationally expensive. The proposed methodology is not restricted to guided-wave SHM and is applicable to a broad class of inverse problems in computational mechanics where fidelity hierarchies are available.

Chapter 2

Background and Literature Review

This chapter establishes the technical foundation for the multi-fidelity surrogate framework, progressing from inverse problem theory through physics-based modeling to machine learning approaches.

2.1 Inverse Problems in Structural Health Monitoring: Formulation and Challenges

Structural health monitoring fundamentally deals with inference. The forward problem involves predicting structural response given properties, geometry, and loading conditions through physics-based models discretized using methods such as finite elements. The inverse problem reverses this: given measured responses, what can be inferred about unknown structural properties, including damage presence and extent.

Consider a structural system characterized by parameter vector $\boldsymbol{\theta}$ representing damage attributes such as location, depth, and width. The forward model $\mathcal{M}$ maps these parameters to predicted response $\mathbf{y}$:

$$\mathbf{y} = \mathcal{M}(\boldsymbol{\theta}) \tag{2.1}$$

In practice, the observed response $\mathbf{y}_{\text{obs}}$ differs from the model prediction due to measurement noise and modeling errors, and is represented as

$$\mathbf{y}_{\text{obs}} = \mathcal{M}(\boldsymbol{\theta}) + \boldsymbol{\epsilon}, \tag{2.2}$$

where $\boldsymbol{\epsilon}$ denotes a stochastic error term that is commonly modeled as a Gaussian random variable with zero mean and specified covariance. The inverse problem consists of inferring the unknown parameter vector $\boldsymbol{\theta}$ from the observations $\mathbf{y}_{\text{obs}}$.

Inverse problems in damage identification are typically ill-posed. Small changes in measured response can produce large changes in estimated damage parameters, while multiple damage configurations may generate indistinguishable responses within measurement precision. Hadamard defined well-posed problems as having solutions that exist, are unique, and depend continuously on data. Damage identification often violates uniqueness, e.g. a shallow wide crack might produce the same response as a deep narrow one.

Classical deterministic methods treat damage identification as optimization, minimizing mismatch between predicted and measured responses. Khatir et al. [2017] demonstrated isogeometric analysis with teaching-learning-based optimization for beam and plate damage identification, employing two-stage frameworks: damage localization using modal strain energy ratios followed by severity estimation via optimization. These methods produce point estimates without indicating when multiple solutions exist, limiting practical utility since decision-makers cannot assess result confidence.

Bayesian inference naturally accommodates ill-posedness by computing probability distributions over all plausible damage configurations. The posterior distribution

$$p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}}) \propto p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta}), \tag{2.3}$$

where $p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})$ is the likelihood and $p(\boldsymbol{\theta})$ is the prior, encodes what measurements reveal about unknown parameters, including correlations and ranges of consistent values. Simon et al. [2024] demonstrated that Bayesian methods address locally identifiable problems by explicitly accounting for environmental influences. Huang et al. [2018] documented the evolution from deterministic model updating to sparse Bayesian learning over two decades. Multi-modal posteriors naturally arise when data cannot distinguish between damage scenarios.

Multimodal posteriors and computational requirements

Real damage identification problems frequently produce multi-modal posteriors. Jesus et al. [2018] reported bimodal distributions in cable system identification for the Tamar Bridge, corresponding to the two main cables—a physically meaningful feature. For guided waves, Cantero-Chinchilla et al. [2019] developed Bayesian methodology eliminating arbitrary choices in time-frequency transform selection, while Zhao et al. [2020] applied Gibbs sampling for multi-damage localization with full uncertainty quantification.

The main computational challenge is that evaluating posteriors requires repeated forward model evaluations. MCMC algorithms may need tens of thousands of forward simulations. If each evaluation requires minutes to hours, direct Bayesian inversion becomes impractical.

2.2 Guided-Wave Propagation in Thin Structures: Physics-Based Modeling Approaches

2.2.1 Lamb wave fundamentals

Lamb waves are elastic guided waves propagating in thin plate-like structures with two parallel free surfaces. The wave field consists of symmetric and antisymmetric modes, denoted $S_0, S_1, \ldots$ and $A_0, A_1, \ldots$, respectively. At low frequency-thickness products, only fundamental modes S_0 and A_0 dominate the wave propagation. The S_0 mode involves predominantly in-plane particle motion, while A_0 involves predominantly out-of-plane flexural motion as shown in Figure 2.1.

When Lamb waves encounter damage such as notches or cracks, they undergo scattering, reflection, and mode conversion depending on damage geometry, wave frequency, and incidence angle. Soleimanpour and Ng [2016] showed that A_0 mode scattering characteristics are strongly influenced by notch size and orientation. Lamb wave responses contain detailed damage information, but interpretation requires sound physical basis.

Fidelity hierarchy in physics-based models

Physics-based models for guided-wave propagation span a spectrum of fidelity levels. For long thin panels, two-dimensional plane strain finite element models offer higher fidelity by explicitly resolving through-thickness stress and displacement fields while maintaining tractable computational cost. These models typically require 10–20 elements per wavelength, see Moser et al. [1999] and time steps satisfying the Courant-Friedrichs-Lewy (CFL) stability criterion, resulting in computation times of 30 minutes to over an hour per simulation for damaged beam models with tens of thousands of elements and lakhs of time steps.

One-dimensional beam theories reduce computational cost by assuming prescribed displacement fields through the thickness. Classical theories like Euler-Bernoulli and Timoshenko provide efficient approximations but have limited ability to represent asymmetric damage features. Refined theories incorporating zigzag displacement patterns can capture more complex deformation modes while retaining computational efficiency orders of magnitude faster than two-dimensional models.

This trade-off between accuracy and computational cost motivates multi-fidelity strategies that combine models of different resolution to achieve high-fidelity predictions at reduced cost.

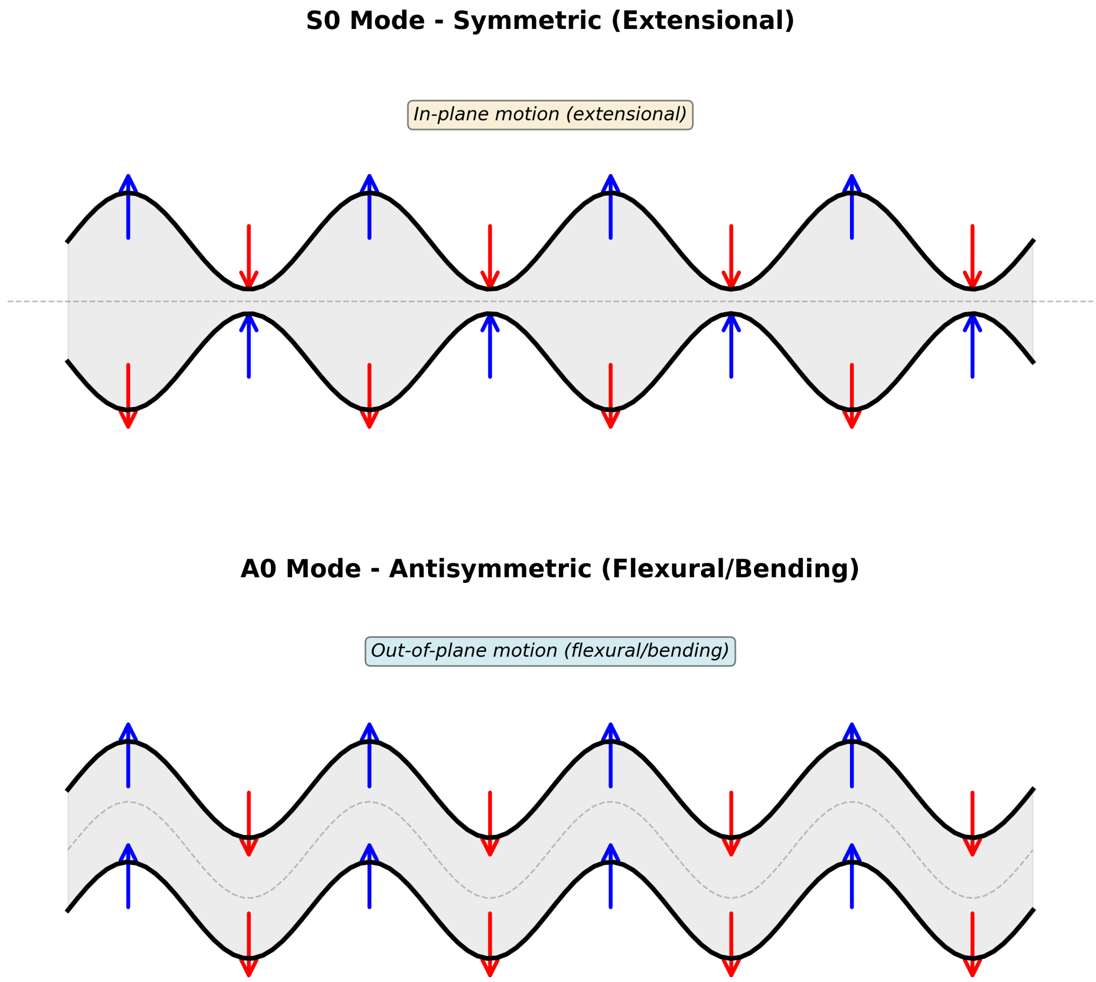


Figure 2.1: Figure illustrating propagation of A_0 (antisymmetric) and S_0 (symmetric) Lamb-wave modes.

2.3 Machine Learning Surrogates for Structural Health Monitoring

Surrogate models approximate the input-output relationship of computationally expensive numerical models. Instead of running finite element simulations for every candidate damage parameter set, surrogates provide fast approximations of corresponding structural responses. Training occurs offline; once trained, surrogate evaluations take milliseconds rather than minutes.

2.3.1 Gaussian process regression

Gaussian process regression provides probabilistic surrogates with built-in uncertainty quantification. By defining prior distributions over functions and conditioning on observed training data, Gaussian processes yield posterior predictive distributions expressing predictions through both mean response and associated variance.

Worden and Cross [2018] applied Gaussian process models to the Z24 bridge benchmark, modeling temperature effects by dividing structural response into distinct regimes and switching between response surfaces as environmental conditions changed. Möller et al. [2025] developed grey-box Gaussian process models embedding physics-informed priors into covariance structure to improve damage detection under climate variability.

The primary limitation is computational scalability. Standard implementations require $\mathcal{O}(n^3)$ operations, where n is the number of training samples, restricting practical use to moderate dataset sizes.

2.3.2 Neural network approaches

Neural networks dominate recent surrogate modeling developments. Dadras and Huang [2022] classified neural network surrogates into parametric models where damage parameters serve as inputs, and nonparametric models where measured signals map directly to outputs. Pandey et al. [2022] reported near-perfect accuracy using explainable one-dimensional convolutional neural networks for Lamb wave damage detection in aluminum plates.

Wave responses are inherently high-dimensional. A single time-domain signal may contain approximately 1500 data points, making direct surrogate modeling inefficient and prone to overfitting.

Autoencoders address this by learning compressed latent representations. An encoder maps high-dimensional input signals to low-dimensional latent space, while a decoder reconstructs the original signal. Li et al. [2024] demonstrated mechanics-informed autoencoders achieving up to 35% improvement over standard autoencoders. Pakzad et al. [2025] combined variational autoencoders with one-class support vector machines and isolation forest for bridge damage detection under environmental variability. Wu et al. [2025] demonstrated 1D-CNN denoising autoencoders improving signal-to-noise ratio by 30.63 dB for ultrasonic guided wave pipeline defect detection.

Ensemble learning methods have also shown strong performance. Dong et al. [2020] achieved coefficients of determination exceeding $R^2 = 0.99$ using gradient boosting techniques like XGBoost for problems ranging from concrete electrical resistivity prediction to wind turbine foundation assessment.

Integration of autoencoders with ensemble models is emerging. Prayogo and Yang [2025] combined variational autoencoders with XGBoost and CatBoost for structural damage classification. However, applications to wave response prediction remain largely unexplored.

2.3.3 Uncertainty quantification in surrogate models

Deterministic neural surrogates provide point estimates without confidence bounds, problematic for inverse inference where surrogate uncertainty should propagate to damage parameter posteriors. Bayesian neural networks address this by placing distributions over network weights. Tripathy and Bilionis [2018] developed deep uncertainty quantification approaches for surrogate modeling, while Vega and Todd [2022] applied variational Bayesian neural networks to miter gate condition monitoring. However, most deployed surrogates remain deterministic.

2.4 Multi-Fidelity Learning: Methods and Applications

Multi-fidelity modeling provides a principled framework for combining models of differing accuracy and computational cost. The foundational work of Kennedy and O'Hagan [2000] introduced a Bayesian formulation in which the high-fidelity response is expressed as a scaled low-fidelity model augmented by a discrepancy term. This framework has been successfully applied to model calibration and uncertainty quantification problems Absi and Mahadevan [2016].

However, Gaussian process–based multi-fidelity models inherit scalability limitations and become impractical for large datasets or high-dimensional outputs. To overcome these challenges, neural network–based multi-fidelity strategies have been proposed. Composite neural architectures with explicitly coupled low- and high-fidelity subnetworks improve scalability while retaining the ability to learn cross-fidelity correlations Meng and Karniadakis [2020].

Transfer learning offers an alternative paradigm for multi-fidelity fusion. In this approach, surrogate models are first trained using abundant low-fidelity data and subsequently fine-tuned using a limited number of high-fidelity samples, thereby reducing the cost of high-fidelity data acquisition, see De et al. [2020]. For problems involving high-dimensional outputs, latent-space multi-fidelity strategies have been adopted, where dimensionality reduction is performed prior to multi-fidelity correction.

In the context of wave-based structural response modeling, several recent studies have used multi-fidelity and transfer learning strategies to reduce simulation–experiment discrepancy. For example, Nerlikar et al. [2024] proposed a physics-embedded deep-learning digital twin to improve forward prediction of guided-wave time-series responses. However, these works remain focused on forward modeling and do not address inverse parameter estimation.

Deep learning has been widely applied to damage diagnosis using wave-based structural response data. In many studies, the learning task is formulated as a classification problem, distinguishing pristine and damaged states or among discrete damage categories. Rai and Mitra [2021] proposed a multi-headed one-dimensional convolutional neural network (1D-CNN) for Lamb-wave–based damage detection, trained using a combination of finite element–generated and experimental data. The numerical data in this study were generated using two-dimensional plane strain finite element models. Subsequently, Rai and Mitra [2022] extended this framework by introducing a transfer learning strategy in which a pre-trained autoencoder serves as a source model whose learned representations are transferred to a CNN classifier, thereby reducing retraining requirements under changed inspection conditions.

In these works, damage-sensitive features are extracted through baseline subtraction between pristine and damaged signals, and the learning objective remains classification between pristine and damaged states. While effective for detection tasks, such formulations do not address inverse damage characterization, where the objective is to estimate continuous damage parameters along with their associated uncertainty.

More recently, Abhishek [2025] investigated joint learning and transfer learning frameworks for notch localization and depth estimation by mapping low-fidelity time-domain simulations to high-fidelity finite element or experimental responses. His primary contribution lies in the generation and use of multi-source data, namely, one-dimensional guided-wave responses obtained from spectral finite element formulations based on zigzag theory (Jain and Kapuria [2023, 2024a]), two-dimensional finite element simulations (implemented in commercial solver ABAQUS), and a limited set of experimental measurements. However, the neural network architectures themselves are largely adopted from existing literature. The forward modeling component relies on one-dimensional convolutional neural network (1D-CNN) autoencoder architectures originally proposed by Nerlikar et al. [2024], while the inverse mapping strategy resembles regression-based adaptations of CNN frameworks previously used for Lamb-wave damage classification by Rai and Mitra [2022]. The architectural modifications introduced are incremental in nature, consisting primarily of parameter tuning and adjustments to network depth or training strategy, rather than the development of a fundamentally new multi-fidelity learning architecture.

Moreover, the framework does not introduce a low-fidelity surrogate model in the sense required for a multi-fidelity surrogate model (Chakraborty [2021]). This distinction becomes critical for Bayesian inverse problems, where thousands of forward model evaluations are required during Markov Chain Monte Carlo (MCMC) sampling, and hence there is a need for extremely cheap forward evaluations. In his work, physics-based low-fidelity models such as zigzag theory and spectral finite element formulations are used exclusively for data generation, and are not converted into surrogate models for repeated forward evaluation. As a result, the learned mapping operates as a deterministic regression model, and the inverse problem is solved in a point-estimation setting without uncertainty quantification, thereby not investigating the possibility of multiple solutions of damage characterization. Surrogate approximation errors arising from limited high-fidelity data are not explicitly modeled, and no probabilistic likelihood is constructed. As a result, uncertainty quantification and posterior inference are not addressed.

Furthermore, the parametric scope of the study is limited. Notch width is not treated as an independent damage parameter, and parametric variation is performed only along a subset of damage variables, often one parameter at a time. No systematic sensitivity analysis is conducted to assess parameter identifiability or to quantify the relative influence of different damage parameters on the measured response. Importantly, the effect of high-fidelity data availability on surrogate performance is not investigated; the framework does not examine how prediction accuracy or inverse estimates evolve as the number of high-fidelity training samples is varied.

2.5 Bayesian Inference and Uncertainty Quantification

Bayesian inference with surrogate models requires careful treatment of likelihood construction, prior specification, and posterior computation. Standard likelihood for structural inverse problems assumes Gaussian measurement errors:

$$\log L(\theta \mid y_{\text{obs}}) \propto -\frac{1}{2\sigma^2}\|y_{\text{pred}}(\theta) - y_{\text{obs}}\|^2 \tag{2.4}$$

Kennedy and O'Hagan [2001] proposed including model inadequacy through Gaussian process discrepancy terms, enabling joint inference of calibration parameters and model error.

MCMC methods sample posteriors by constructing Markov chains with stationary distributions equal to the posterior. Haario et al. [2001, 2006] developed Delayed Rejection Adaptive Metropolis (DRAM), adapting proposal covariance from chain history with delayed rejection for difficult proposals. Ching and Chen [2007] introduced Transitional MCMC for structural engineering, using temperature schedules to bridge prior and posterior for multimodal and peaked distributions.

Population-based samplers address tuning challenges using multiple chains simultaneously. ter Braak [2006] developed Differential Evolution Markov Chain (DE-MC), proposing jumps as multiples of differences between random parameter vectors. Vrugt et al. [2009] extended this to DREAM with subspace sampling for high-dimensional problems. Differential evolution provides globally optimized starting points improving convergence. Reiser et al. [2025] established that ignoring surrogate uncertainty leads to biased and overconfident posteriors, proposing marginalization and joint inference approaches. For multi-fidelity contexts, neural network surrogates with calibrated epistemic uncertainty propagated through Bayesian inference remain underdeveloped.

2.6 Global Sensitivity Analysis Using Variance Decomposition

Suppose a computational model generates predictions that depend on several input parameters. Before solving inverse problems to estimate these parameters, a natural question arises to inspect which parameters matter most. If the model has ten inputs but only two significantly influence the output, attempting to infer all ten from noisy measurements wastes computational effort and introduces unnecessary uncertainty. This question motivates sensitivity analysis.

Local sensitivity methods compute derivatives of the output with respect to each input at a specific parameter value. A large derivative indicates a strong local influence. However, derivatives depend on the evaluation point and say nothing about parameter importance when inputs vary across their entire ranges. If a parameter has high local sensitivity in one region but low sensitivity elsewhere, a derivative computed at a single point gives an incomplete picture.

Global sensitivity analysis addresses this limitation by quantifying parameter importance across the full parameter space rather than at isolated points, see Saltelli et al. [2008]. The focus shifts from local sensitivity at a specific point to quantifying how much of the overall output variability (across all plausible parameter values) is attributable to each input parameter. Variance-based methods do this by decomposing the total output variance into contributions from individual parameters and their interactions.

2.6.1 Variance decomposition and Sobol indices

Consider the computational model 2.1 that maps input parameters $\boldsymbol{\theta} \in \mathbb{R}^d$ to output Y. The inputs are treated as independent random variables with specified probability distributions. The output Y is consequently random. Variance-based sensitivity analysis decomposes the variance of Y to assign importance to each input and to interactions among inputs.

The foundational decomposition, introduced by Sobol [1993], expresses the model as a sum of functions with increasing dimensionality. For two parameters θ_1 and θ_2, the model can be written

$$\mathcal{M}(\theta_1, \theta_2) = f_0 + f_1(\theta_1) + f_2(\theta_2) + f_{12}(\theta_1, \theta_2), \tag{2.5}$$

where f_0 is a constant, f_1 and f_2 represent main effects of individual parameters, and f_{12} captures the interaction between them. This decomposition extends to arbitrary dimension with unique terms satisfying orthogonality conditions. Taking variances of each term yields a corresponding decomposition of the total output variance

$$\mathrm{Var}(Y) = V_1 + V_2 + V_{12}, \tag{2.6}$$

where $V_1 = \mathrm{Var}[f_1(\theta_1)]$ represents variance contribution from θ_1 alone, and so forth.

Sobol indices are the normalized variance contributions. The first-order Sobol index for parameter i is defined as

$$S_i = \frac{V_i}{\mathrm{Var}(Y)}, \tag{2.7}$$

which measures the fraction of output variance attributable to parameter i acting independently. A value $S_i = 0.7$ means that if all other parameters were fixed at any values and only θ_i varied according to its distribution, 70% of the output variance would remain. This index isolates the main effect of θ_i without accounting for how it interacts with other parameters.

Interactions complicate the picture. A parameter might have small main effect but contribute significantly through interactions with others. The total-effect Sobol index, introduced by Homma and Saltelli [1996], accounts for all contributions involving parameter i, including interactions:

$$S_{T_i} = S_i + \sum_{j \neq i} S_{ij} + \sum_{j \neq i, k \neq i, j < k} S_{ijk} + \dots \tag{2.8}$$

This index measures the total fraction of variance that involves parameter i in any capacity. The difference $S_{T_i} - S_i$ quantifies the contribution from interactions. If $S_{T_i} \approx S_i$, the parameter acts mostly independently. If $S_{T_i} \gg S_i$, interactions dominate its influence.

2.6.2 Monte Carlo estimation methods

The integrals defining Sobol indices cannot be evaluated analytically for most computational models. Monte Carlo estimation provides a practical approach. The first-order index can be expressed as

$$S_i = \frac{\mathrm{Var}_{\theta_i}\left[\mathbb{E}_{\boldsymbol{\theta}_{\sim i}}(Y|\theta_i)\right]}{\mathrm{Var}(Y)}, \tag{2.9}$$

where $\mathbb{E}_{\boldsymbol{\theta}_{\sim i}}(Y|\theta_i)$ denotes the expected output when θ_i is fixed and all other parameters vary. The outer variance operator measures how this conditional expectation varies as θ_i changes. Similarly, the total-effect index is

$$S_{T_i} = \frac{\mathbb{E}_{\boldsymbol{\theta}_{\sim i}}\left[\mathrm{Var}_{\theta_i}(Y|\boldsymbol{\theta}_{\sim i})\right]}{\mathrm{Var}(Y)}, \tag{2.10}$$

representing the expected variance remaining when all parameters except θ_i are fixed.

Direct Monte Carlo evaluation of these conditional expectations requires nested sampling loops, which rapidly becomes computationally prohibitive. Saltelli proposed an efficient sampling scheme that circumvents nested loops. The method generates two independent sample matrices $\mathbf{A}$ and $\mathbf{B}$, each of size $N \times d$, where each row represents one parameter set sampled from the joint distribution. For each parameter i, a hybrid matrix $\mathbf{AB}_i$ is constructed by copying $\mathbf{A}$ and replacing its i-th column with the corresponding column from $\mathbf{B}$. The model is evaluated at all sample points to obtain response vectors $\mathbf{Y}_A$, $\mathbf{Y}_B$, and $\mathbf{Y}_{AB_i}$ for $i = 1, \ldots, d$.

Using these evaluations, the first-order index is estimated as

$$S_i = \frac{\frac{1}{N}\sum_{j=1}^{N} Y_B^{(j)}\left(Y_{AB_i}^{(j)} - Y_A^{(j)}\right)}{\mathrm{Var}(Y)}, \tag{2.11}$$

and the total-effect index as

$$S_{T_i} = \frac{\frac{1}{2N}\sum_{j=1}^{N}\left(Y_A^{(j)} - Y_{AB_i}^{(j)}\right)^2}{\mathrm{Var}(Y)}. \tag{2.12}$$

Computing indices for all d parameters requires $N(d+2)$ model evaluations, see Saltelli [2002]. When N is chosen to be several thousand, this becomes expensive for models requiring substantial computational time per evaluation. Surrogate models that approximate the original model with negligible evaluation cost make Sobol analysis tractable for such cases, [Saltelli et al., 2008].

Many computational models produce time-dependent or spatially distributed outputs rather than scalar responses. A dynamic simulation generates a time series $Y(t)$ for $t \in [0, T]$. A spatial model produces a field $Y(\mathbf{x})$ over a domain. The standard Sobol indices defined for scalar outputs do not directly apply.

The most straightforward extension computes Sobol indices pointwise. For a time series, indices are calculated separately at each time instant:

$$S_i(t) = \frac{\mathrm{Var}_{\theta_i}\left[\mathbb{E}_{\boldsymbol{\theta}_{\sim i}}(Y(t)|\theta_i)\right]}{\mathrm{Var}(Y(t))}. \tag{2.13}$$

This produces time-varying sensitivity profiles revealing which parameters dominate at different stages of the process. Early-time sensitivity may differ from late-time behavior. Pointwise indices are computed independently at each time step using the same sample matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{AB}_i$, with model evaluations providing response time series rather than scalars.

For applications requiring a single global importance measure across time, pointwise indices can be aggregated. Common approaches include averaging indices over time, integrating them against a weight function that emphasizes critical time windows, or extracting maximum values to identify peak sensitivity. The choice depends on whether early-time, late-time, or transient behavior matters most for the application. In cases where the entire time evolution matters equally, simple averaging provides a reasonable summary.

Chapter 3

Multi-Fidelity Surrogate Framework with Uncertainty Quantification

3.1 Methodology Overview and Design Philosophy

This chapter presents the methodology developed in this thesis for constructing an uncertainty-aware multi-fidelity surrogate model for guided-wave response prediction and Bayesian inverse damage characterization.

The central challenge addressed is the prohibitive computational cost of high-fidelity guided-wave simulations when embedded within inverse inference frameworks that require repeated forward evaluations. While high-fidelity (HF) models accurately resolve wave-damage interactions, only a limited number of HF simulations can be generated in practice. This data scarcity leads to unavoidable surrogate approximation errors that must be explicitly accounted for to avoid overconfident inverse estimates.

The proposed framework combines:

- physics-based low- and high-fidelity forward models,
- latent-space dimensionality reduction using an autoencoder trained on abundant low-fidelity data,
- parameter-to-latent regression using a data-efficient learning model,
- transfer learning to incorporate high-fidelity information, and
- probabilistic modeling of surrogate-induced uncertainty.

Uncertainty in this work arises primarily due to limited high-fidelity training data and the resulting surrogate approximation error. Environmental and operational variability are not considered.

3.2 Fidelity Hierarchy: Forward Model Definitions

3.2.1 High-Fidelity model: Two-dimensional plane strain Finite Element Analysis

The high-fidelity forward model resolves guided-wave propagation and wave–damage interaction using a two-dimensional plane strain finite element formulation. This assumption is appropriate for long, thin panel-like structures commonly encountered in aerospace applications, where deformation along the width direction is constrained and strain variation is negligible, as used by Pandey et al. [2022]. The damage is modeled as a rectangular notch (extending along the panel width), parameterized by its location, width, and depth. Under these conditions, the plane strain formulation provides a physically consistent representation of a long three-dimensional crack.

Geometry and material properties

Figure 3.1 illustrates the beam geometry considered in this study. The structure is a homogeneous aluminum beam with total length $L = 3$ m and thickness $h = 1.5$ mm. The thickness of 1.5 mm is common in airplanes and ships; see Gopalakrishnan et al. [2007]. The material properties correspond to standard aluminum alloy: Young's modulus $E = 70$ GPa, mass density $\rho = 2700$ kg/m^3, and Poisson's ratio $\nu = 0.33$. The beam is modeled under free-free boundary conditions, representing an unconstrained panel segment.

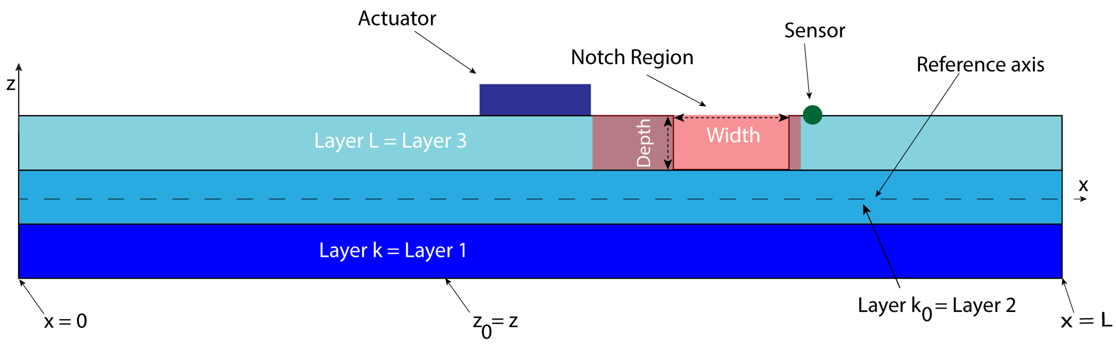


Figure 3.1: Schematic of the aluminum beam geometry showing actuator placement at $x = 1.65$, sensing at $x = 1.95$ m, and the permissible notch location range. The beam has length $L = 3$ m and thickness $h = 1.5$ mm under free-free boundary conditions.

The rectangular notch simulates simplified localized damage such as fatigue cracking or corrosion-induced material loss. Three parameters define the notch geometry: the location x_d measured from the left boundary to the notch center, the width w_d representing the notch extent along the beam axis, and the depth d_d indicating the penetration through the beam thickness from the top surface. The notch is implemented by removing elements within the notch region from the structural domain, creating a physical discontinuity that interacts with propagating guided waves.

The wave propagation setup employs a pitch–catch configuration with a single actuator paired with a sensor at location $x = 1.95m$. The actuator is positioned at $x = 1.65$ m from the left boundary on the top surface of the beam and spans a width of 0.7 mm. This location places the actuator approximately at the mid-span region, allowing sufficient propagation distance for wave development before interacting with the damage zone and to avoid boundary reflection. This arrangement positions the notch region between the actuator and sensor, ensuring that scattered and transmitted wave components carry damage-related information to the measurement location.

The region highlighted in Figure 3.1 between actuator and sensor region spans 0.181 m and represents the permissible range for the notch center location, which varies from $x_d = 1.6545$ m to $x_d = 1.8355$ m along the beam axis. The notch depth ranges from $d_d = 0.1$ mm to $d_d = 1.0$ mm, corresponding to approximately 7% to 67% of the beam thickness. The notch width varies from $w_d = 0.1$ mm to $w_d = 1.2$ mm. These parameter ranges—notch location, notch width, notch depth, and receiver location to the right of the actuator—define the four-dimensional input space over which the surrogate model is trained and subsequently used for inverse inference.

Actuation signal characteristics

The actuator generates a Hanning-windowed sinusoidal burst at a center frequency of $f_c = 100$ kHz. The excitation signal is defined as

$$F(t) = \begin{cases} \frac{1}{2}\left[1 - \cos\left(\frac{2\pi t}{T_a}\right)\right]\sin(2\pi f_c t), & 0 \leq t \leq T_a \\ 0, & t > T_a \end{cases} \tag{3.1}$$

where $T_a = 50$ μs is the active excitation duration. The Hanning window modulation produces a narrowband signal with smooth onset and decay, minimizing spectral leakage and enabling clear identification of wave packets in the time-domain response. At 100 kHz central frequency, the excitation generates both fundamental symmetric (S_0) and antisymmetric (A_0) Lamb wave modes in the thin aluminum plate. The chosen frequency lies within a regime where these modes propagate with distinct group velocities, facilitating mode identification in the received signals. Refer to Figure 3.2 for the visualization.

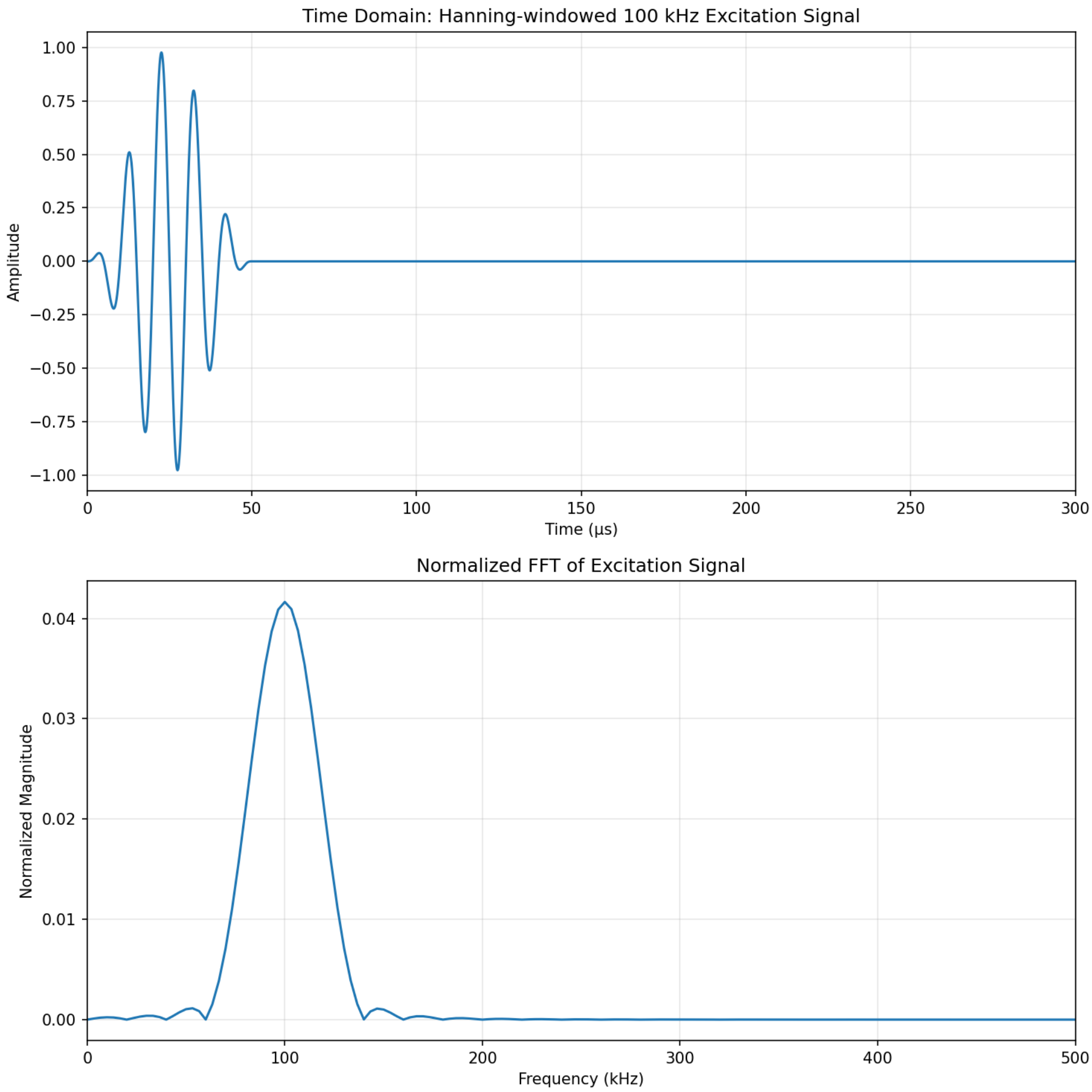


Figure 3.2: Hanning-windowed tone burst excitation signal at center frequency $f_c = 100$ kHz with active duration $T_a = 50$ μs (top) and corresponding frequency spectrum showing narrowband energy concentration (bottom).

Spatial and temporal discretization

The spatial domain is discretized using 8-noded quadrilateral elements under plane strain conditions. A mesh of 10,000 elements along the axial direction and 10 elements through the thickness is employed, giving a total of 100,000 elements. Field variables are interpolated using Lagrange shape functions. Under plane strain conditions, the out-of-plane direction is constrained such that deformation is suppressed; consequently, the loading in that direction (the y-direction in this work) must be either zero or spatially uniform. This mesh density ensures adequate resolution for both wavelength representation and damage geometry. For the A_0 mode propagating at 100 kHz in 1.5 mm thick aluminum (wavelength approximately 17 mm), the axial discretization provides approximately 30 elements per wavelength.

The through-thickness discretization with 10 elements captures the characteristic displacement variations of Lamb wave propagation. To resolve the smallest notch configuration (0.1 mm $\times$ 0.1 mm), local mesh refinement was implemented in the vicinity of the notch, achieving a minimum inter-nodal distance of 0.05 mm. This fine discretization directly influences the timestep size and computational time required for the time integration scheme.

The governing equations obtained after spatial discretization of the continuum using the finite element method take the form of a system of second-order ordinary differential equations in time, commonly referred to as the equations of motion:

$$\mathbf{M}\,\ddot{\mathbf{u}}(t) + \mathbf{C}\,\dot{\mathbf{u}}(t) + \mathbf{K}\,\mathbf{u}(t) = \mathbf{f}(t), \tag{3.2}$$

where $\mathbf{u}(t)$ is the vector of nodal displacement degrees of freedom, $\mathbf{M}$ is the global mass matrix, $\mathbf{K}$ is the global stiffness matrix, and $\mathbf{f}(t)$ denotes the external force vector.

Damping is introduced through the global damping matrix $\mathbf{C}$, which is modeled using standard Rayleigh damping:

$$\mathbf{C} = \alpha\,\mathbf{M} + \beta\,\mathbf{K}, \tag{3.3}$$

where α and β are mass- and stiffness-proportional damping coefficients, respectively; these coefficients are selected to provide light numerical damping over the frequency range of interest, sufficient to suppress nonphysical high-frequency oscillations without significantly affecting wave propagation characteristics.

Time integration of the equations of motion is performed using an explicit central difference (CD) scheme. The total simulation duration is $T = 300$ μs with 150,000 time steps, yielding a time step $\Delta t = 2$ ns. This small time step satisfies the CFL stability condition for the finest mesh elements and highest wave velocities present in the model. Each high-fidelity simulation requires approximately 38 to 40 minutes of computation on a GPU-accelerated platform, implemented in PyTorch. This computational cost, while acceptable for generating training datasets, renders direct use of the high-fidelity model within Bayesian inverse inference impractical when thousands of forward evaluations are required.

3.2.2 Low-Fidelity model: refined zigzag theory for beam structures

The low-fidelity model employs a one-dimensional beam formulation based on Refined Zigzag Theory (RZT), originally developed by Tessler et al. [2007]. The key idea of zigzag theory is the introduction of a fictitious layerwise zigzag function that enriches the axial displacement field, allowing its thickness-wise slope to change from one layer to another while maintaining displacement continuity. This fictitious layer concept enables the representation of piecewise-linear kinematics across the laminate thickness without introducing additional physical layers or full layerwise degrees of freedom. Figure 3.3 illustrates the multilayered beam and the associated zigzag kinematic representation across the thickness.

Classical beam theory limitations for asymmetric notch modeling

Classical one-dimensional beam theories, such as Euler-Bernoulli and Timoshenko, do not explicitly represent the cross-section as a two-dimensional domain. Instead, the cross-sectional geometry is collapsed into a small set of section properties, such as the cross-sectional area $A(x)$, bending stiffness $I(x)$, and shear stiffness $\kappa GA(x)$. As a result, damage in these models is typically introduced through local reductions in these effective properties.

When a surface notch is modeled within Euler-Bernoulli or Timoshenko theory, its effect is represented implicitly by modifying $A(x)$ and $I(x)$ over the damaged region. This representation does not retain information about the through-thickness location of the damage. Consequently, the beam model cannot distinguish between notches located on different surfaces or capture their asymmetric placement relative to the reference axis.

This limitation is particularly relevant for guided-wave propagation, where the interaction between the wavefield and localized geometric discontinuities depends not only on the severity of the damage but also on its location in a cross-section. Classical beam theories therefore provide a reduced representation of notch-like damage that may be insufficient to reproduce certain scattered components of the wave response observed in higher-dimensional continuum models.

Refined zigzag theory: kinematic assumptions

Refined Zigzag Theory overcomes this limitation through piecewise-polynomial displacement fields. The theory employs zigzag functions that capture layer-by-layer variation in axial displacement, refer Figure 3.3. The displacement field takes the form

$$u(x, z, t) = u_0(x, t) - z\frac{\partial w_0}{\partial x}(x, t) + R^{(k)}(z)\,\psi_0(x, t) \tag{3.4}$$

where u_0 is the reference plane axial displacement, w_0 is transverse displacement constant through thickness, and ψ_0 is the shear deformation parameter. The zigzag function $R^{(k)}(z)$ is a piecewise cubic polynomial constructed to satisfy displacement continuity at layer interfaces, shear stress continuity, and traction-free boundary conditions at surfaces. This function distinguishes zigzag theory from Euler–Bernoulli and Timoshenko formulations, which assume linear through-thickness variation.

For a three-layer beam, the zigzag function within each layer is expressed as

$$R^{(k)}(z) = a_0^{(k)} + a_1^{(k)} z + a_2^{(k)} z^2 + a_3^{(k)} z^3 \tag{3.5}$$

The coefficients $a_0^{(k)}$, $a_1^{(k)}$, $a_2^{(k)}$, and $a_3^{(k)}$ associated with the zigzag displacement field in the k-th layer are determined by enforcing continuity and boundary conditions across layer interfaces, see Kapuria and Hagedorn [2007]. Specifically, the following conditions are imposed.

First, displacement continuity is enforced at each interface $z = z_k$ between adjacent layers, ensuring that the axial displacement field remains continuous through the thickness:

$$u^{(k)}(x, z_k, t) = u^{(k+1)}(x, z_k, t). \tag{3.6}$$

Second, continuity of transverse shear stress is enforced at all layer interfaces, which translates into continuity of the corresponding shear strain measures:

$$\tau_{xz}^{(k)}(x, z_k, t) = \tau_{xz}^{(k+1)}(x, z_k, t). \tag{3.7}$$

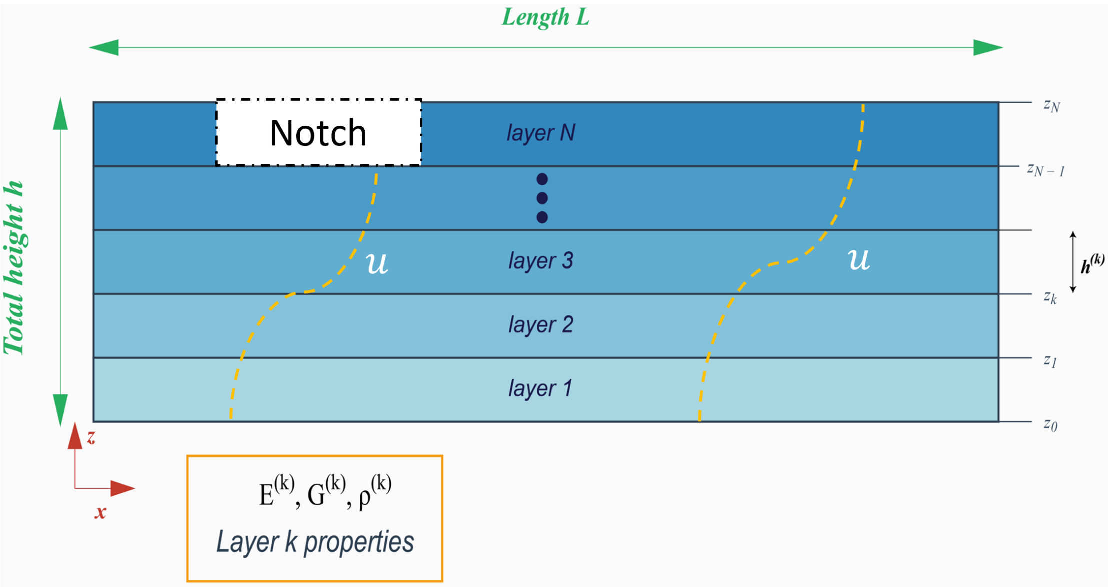


Figure 3.3: Zigzag beam kinematics for an N-layer laminate with a representative notch for demonstration. The yellow dotted curve represents the axial displacement function represented through thickness

Third, the zigzag function is constructed such that its thickness-wise average over the entire laminate vanishes, ensuring that the zigzag contribution does not alter the global kinematics captured by the coarse-scale displacement variables:

$$\int_{-h/2}^{h/2} \psi(z)\,\mathrm{d}z = 0. \tag{3.8}$$

Finally, traction-free boundary conditions are enforced at the top and bottom surfaces of the laminate, requiring the transverse shear stresses to vanish at $z = \pm h/2$:

$$\tau_{xz}(x, \pm h/2, t) = 0. \tag{3.9}$$

Together, these continuity, normalization, and boundary conditions provide a closed system of equations that uniquely determines the coefficients $a_0^{(k)}$, $a_1^{(k)}$, $a_2^{(k)}$, and $a_3^{(k)}$ for each layer, ensuring physically admissible zigzag kinematics across the multilayered beam.

For homogeneous materials, the original zigzag theory reduces to Timoshenko theory since there are no property discontinuities to generate zigzag behavior. However, by introducing fictitious layers with perturbed properties, one can recover zigzag kinematics even in nominally homogeneous beams. In the presence of a notch, the damaged region naturally exhibits different effective properties than the undamaged material. Treating the notched region as a virtual layer with reduced thickness enables the zigzag formulation to capture asymmetric deformation patterns, see Jain and Kapuria [2024b]

Low-fidelity model configuration and loading

The one-dimensional model considers the same aluminum beam geometry as the high-fidelity model: length $L = 3$ m, thickness $h = 1.5$ mm, with material properties $E = 70$ GPa, $\rho = 2700$ kg/m^3, and $\nu = 0.33$. The beam is divided into three fictitious layers through the thickness to enable zigzag kinematics. Free-free boundary conditions are applied at both ends. The spatial domain is discretized using 2000 elements, and the simulation is performed over a total duration of $T = 300$ μs with a time step of $\Delta t = 65$ ns. A single zigzag theory simulation completes in approximately 20 seconds on a GPU-accelerated platform, compared to 38 minutes for the two-dimensional finite element model. This speedup factor exceeding 100 enables the generation of large low-fidelity datasets within practical time constraints. The framework exploits this asymmetry: hundreds of low-fidelity samples can be generated in the time required for a handful of high-fidelity simulations.

The actuator location and excitation signal match the high-fidelity model. However, in the one-dimensional formulation, there is no physical top surface on which to apply the excitation; rather everything is modelled about the neutral axis.

The force is therefore shifted to the neutral axis, which introduces an additional moment to account for the offset. Figure 3.4 illustrates this equivalent loading representation for the 1D beam model.

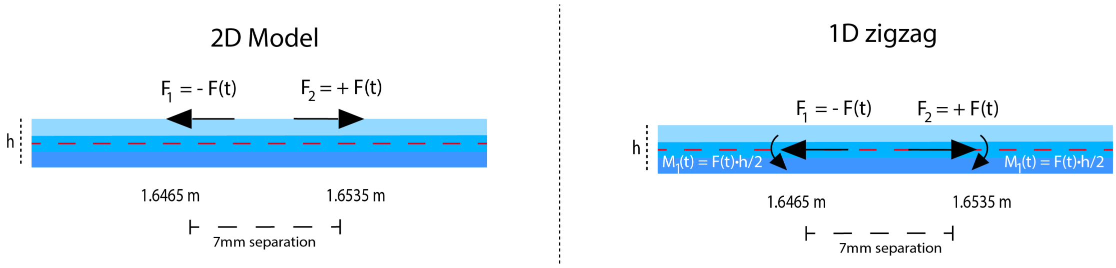


Figure 3.4: Equivalent loading for the one-dimensional zigzag model. The surface-applied force $F(t)$ is transferred to the neutral axis with an accompanying moment $M(t) = F(t) \cdot h/2$ to preserve static equivalence.

Comparison with Euler-Bernoulli and Timoshenko theories

Figure 3.5 demonstrates the accuracy of zigzag beam theory compared to classical beam theories for a damaged beam configuration. The first prominent wave packet on the left corresponds to the symmetric wave mode (S_0), while the last wave packet on the right corresponds to the antisymmetric wave mode (A_0). The oscillatory response observed between these two packets arises from the presence of damage along the propagation path, which induces wave scattering and mode conversion. This intermediate wave component is not captured by Euler–Bernoulli or Timoshenko theories, whereas zigzag theory reproduces it in close agreement with the two-dimensional continuum model.

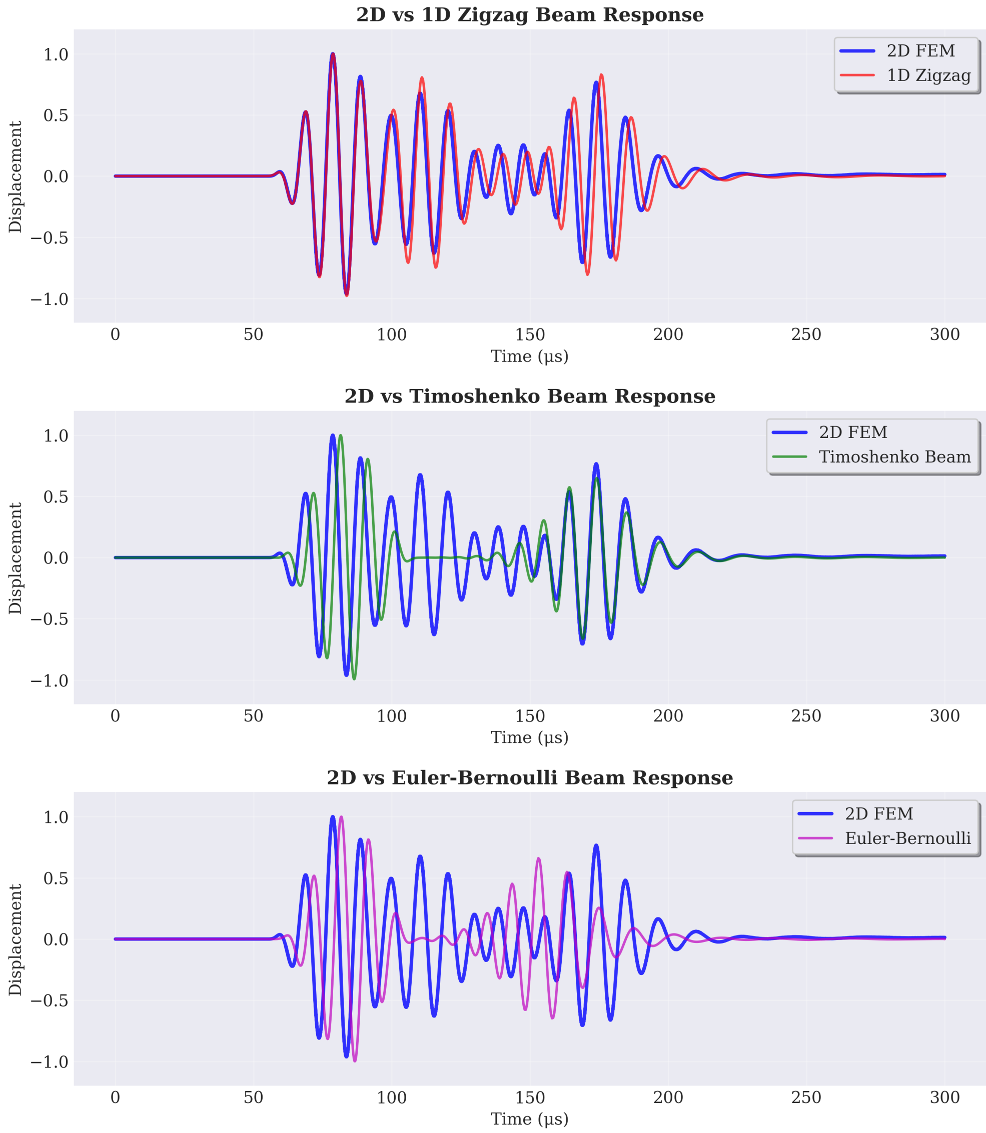


Figure 3.5: Time-domain axial displacement responses from one-dimensional beam theories compared against two-dimensional plane strain FEM. The zigzag formulation captures the scattered wave component between the S_0 and A_0 mode packets, which classical theories fail to reproduce. All responses normalized to unit maximum amplitude.

3.3 Damage Parameterization and Data Generation

The notch—mimicking damage— is characterized in terms of a four-dimensional parameter vector

$$\boldsymbol{\theta} = [x_d,\ w_d,\ d_d]^\top, \tag{3.10}$$

where x_d denotes the axial location of the notch center measured from the actuator, w_d is the notch width, and d_d is the notch depth measured from the top surface.

To construct training and testing datasets in a systematic manner, each parameter is discretized over a prescribed set of values. The resulting datasets are generated by forming Cartesian products of these discrete sets, ensuring structured coverage of the parameter space.

3.3.1 Low-fidelity training dataset

For the low-fidelity (LF) model, the following discretization is adopted: the discrete parameter space was discretized into 12 location values, 12 width values, and 10 depth values, and 12 width values, yielding ($12 \times 12 \times 10$) 1440 unique parameter combinations. The low-fidelity training dataset was chosen to have 750 time response samples, representing 52% of the total parameter space.

The sampling strategy uses a maximin distance criterion to select 750 points from all 1440 possible combinations. This approach maximizes the minimum distance between any two points in the selected set. The algorithm starts with a random initial subset of 750 points and iteratively improves the selection through point swapping. At each iteration, a point is removed and replaced with a candidate from the remaining pool. If the swap increases the minimum pairwise distance, it is accepted. The process continues until no further improvement occurs.

This sampling method ensures uniform coverage of the parameter space and avoids clustering in any region. The resulting sample distribution captures the full range of parameter variations with fewer simulations compared to exhaustive sampling, making it computationally efficient while maintaining representative coverage.

Each simulation produces a guided-wave time-series response consisting of $n_t = 1500$ time samples. This structured dataset provides controlled variation of each damage parameter while maintaining computational tractability, and forms the primary training dataset for the low-fidelity surrogate and autoencoder.

3.3.2 High-fidelity Training and Testing Dataset

The high-fidelity dataset consists of 100 training samples and 50 test samples with no overlap. Selecting only 100 training samples from 1440 possible combinations (approximately 7% coverage) makes obtaining a representative dataset challenging. The maximin distance criterion is applied to ensure maximum spatial coverage across the parameter space.

The 50 test samples are selected from the remaining 1340 combinations using a two-phase strategy. The first phase ensures full coverage by selecting points that represent all 12 location values, all 10 depth values, and all 12 width values at least once. This is achieved through greedy selection where each candidate is scored based on how many uncovered discrete values it contributes. Once full coverage is achieved, the remaining test samples are selected using the maximin distance criterion to maintain uniform spatial distribution. This approach ensures thorough model validation across the entire parameter space.

3.4 Dimensionality Reduction Using Latent-Space Representation

3.4.1 Rationale for latent-space approach

Direct surrogate modeling of guided-wave responses is computationally challenging due to the high dimensionality of the output space. Each simulated response consists of a time-series signal with $n_t = 1500$ samples, and learning a direct mapping from the damage parameter space to such high-dimensional outputs is both data-inefficient and numerically unstable, particularly under limited high-fidelity data availability.

To address this challenge, a latent-space modeling strategy is adopted. The key idea is to first compress the guided-wave responses into a low-dimensional latent representation that captures the dominant response characteristics, and then construct surrogate models in this reduced space. This approach significantly reduces the effective output dimensionality while preserving the information content required for inverse damage characterization.

3.4.2 Autoencoder design and training protocol

Let $\mathbf{u} \in \mathbb{R}^{1500}$ denote a guided-wave response time series. An autoencoder is employed to learn a nonlinear mapping between the response space and a low-dimensional latent space $\mathbf{z} \in \mathbb{R}^m$:

$$\mathbf{z} = \mathrm{Enc}(\mathbf{u}), \qquad \hat{\mathbf{u}} = \mathrm{Dec}(\mathbf{z}), \tag{3.11}$$

where $\mathrm{Enc}(\cdot)$ and $\mathrm{Dec}(\cdot)$ denote the encoder and decoder networks, respectively.

The autoencoder is trained in two stages. Initially, the network is trained on a large dataset of low-fidelity (LF) responses to learn a robust latent representation. Subsequently, the model is fine-tuned on high-fidelity (HF) response data using a reduced learning rate, allowing the learned features to adapt to the characteristics of the HF data while preserving the knowledge acquired from the LF training phase. The architecture, illustrated in Figure 3.6, processes input signals consisting of 1500 datapoints representing the temporal response signal. These inputs are encoded into a compressed latent space of dimension 30, which captures the essential features of the response. The decoder reconstructs the 1500-point response signal from the latent representation. This two-stage training strategy leverages the availability of abundant LF data to establish a well-initialized latent manifold, which is then refined through exposure to HF data to ensure that the learned representation is physically meaningful across both fidelity levels.

3.4.3 Latent dimension selection criterion

The latent dimension m is selected through cross-validation by evaluating the reconstruction accuracy of the autoencoder for $m \in \{10, 20, 30, 40, 50, 60\}$. A latent dimension of $m = 30$ is found to provide a favorable balance between reconstruction fidelity and model complexity. Lower-dimensional representations lead to loss of damage-sensitive features, while higher-dimensional representations offer diminishing returns and increase computational cost. The choice $m = 30$ is therefore adopted throughout this work.

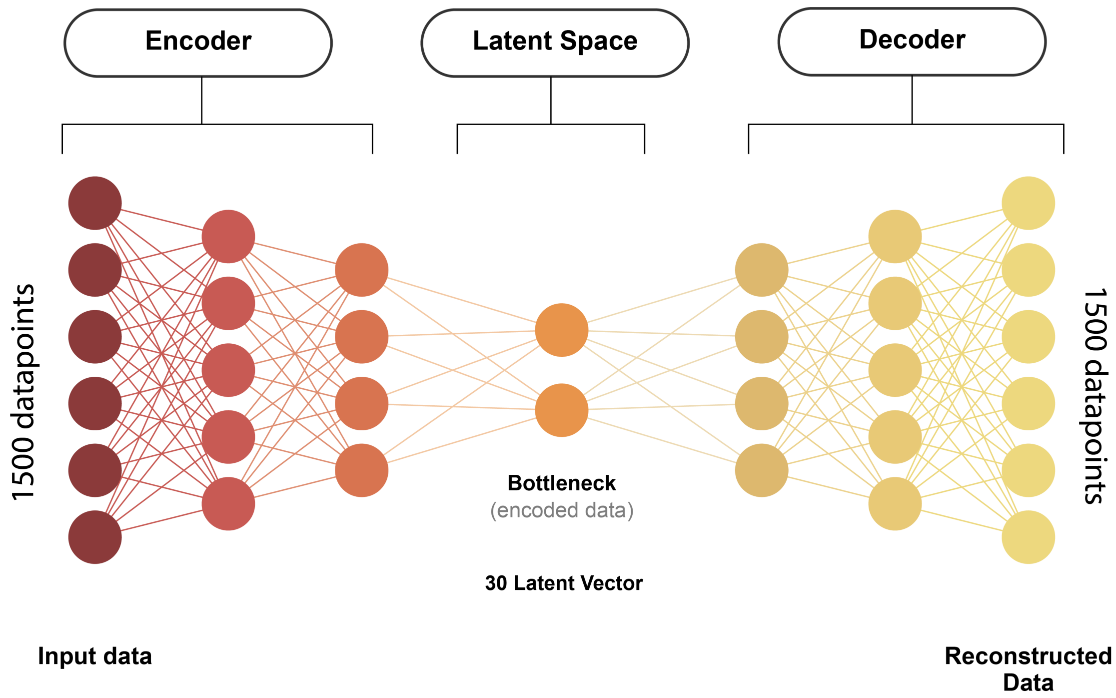


Figure 3.6: Autoencoder architecture for latent-space dimensionality reduction. The encoder compresses 1500-point response signals into a 30-dimensional latent representation, and the decoder reconstructs the response from the latent code.

3.5 Parameter-to-Latent-Space Mapping

Once the latent representation is established, a surrogate model is constructed to map the damage parameter vector $\boldsymbol{\theta}$ to the latent space. This mapping is denoted by

$$\boldsymbol{\mu}_z(\boldsymbol{\theta}) = f(\boldsymbol{\theta}), \tag{3.12}$$

where $f(\cdot)$ provides a deterministic estimate of the latent code corresponding to a given parameter set.

The regression function $f(\cdot)$ is implemented using XGBoost. Empirical evaluation showed that XGBoost consistently outperformed fully connected neural networks in terms of prediction accuracy and robustness for the available dataset. This behavior is attributed to XGBoost's ability to model nonlinear interactions with limited data while maintaining strong regularization properties.

Importantly, the surrogate maps parameters directly to the latent space, and the encoder is not used during inference. The encoder is retained only for constructing latent representations of HF responses during uncertainty quantification, ensuring consistency across fidelity levels.

3.6 Transfer Learning for Multi-Fidelity Integration

High-fidelity information is incorporated through a two-phase transfer learning strategy. In phase one, the autoencoder is trained on the large LF dataset to learn a robust latent representation. Once trained, the autoencoder weights are frozen and the encoder is used to transform the LF responses into the learned latent space of dimension 30. An XGBoost surrogate is then trained to map the damage parameters to this LF latent representation. This is termed as Low Fidelity Surrogate Model (LFSM)

In phase two, the entire autoencoder (both encoder and decoder) is fine-tuned using the limited HF response data under reduced learning rates. Following this fine-tuning, the autoencoder weights are again frozen. A new XGBoost model is then trained both LFSM latent space and the updated latent space obtained from the HF-tuned autoencoder, with more weight given to the new latent space, allowing it to adapt to the high-fidelity physics while maintaining stability through the reduced learning rate strategy. This is what we termed as Multi-Fidelity Surrogate Model (MFSM). The overall architecture combining the autoencoder and XGBoost is illustrated in 3.7.

Alternative approaches were explored during development. These included freezing only the encoder and fine-tuning solely the decoder on HF data, as well as training the XGBoost surrogate directly on the latent space after fine-tuning the autoencoder without subsequent XGBoost fine-tuning. However, both of these approaches resulted in degraded generalization performance, with performance degradation attributed to insufficient HF samples leading to overfitting and inconsistency in the latent representations. The adopted two-phase approach ensures that the latent space structure learned from abundant LF data is preserved while being refined through HF corrections, and the parameter-to-latent mapping maintains physical consistency across fidelity levels.

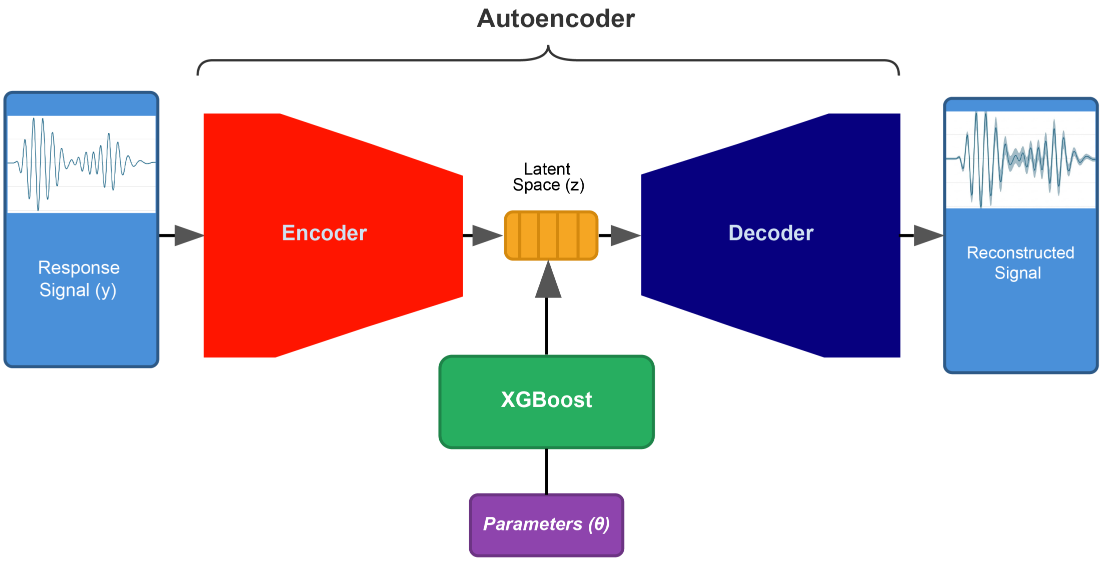


Figure 3.7: Multi-fidelity surrogate modeling framework. (a) Training phase: the autoencoder is pre-trained on 750 low-fidelity responses and fine-tuned on 100 high-fidelity samples where the XGBoost is trained on Low Fidelity latent space and weight adjusted high-fidelity fine-tuned latent space. (b) Inference phase: the XGBoost surrogate maps damage parameters directly to latent space, bypassing the encoder; latent uncertainty is propagated through the decoder to produce uncertainty-aware response predictions.s

3.7 Modeling Surrogate Approximation Uncertainty

Uncertainty in surrogate predictions arises primarily from the limited availability of HF training data, imperfect low-to-high fidelity transfer, and reconstruction errors of the autoencoder. To avoid overconfident inverse estimates, these sources of uncertainty must be explicitly represented prior to Bayesian inference.

Uncertainty is modeled in the latent space by treating the latent variable $\mathbf{z}$ as a Gaussian random vector conditional on the input parameters:

$$\mathbf{z} \mid \boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\mu}_z(\boldsymbol{\theta}), \boldsymbol{\Sigma}_z) . \tag{3.13}$$

The Gaussian assumption is adopted as a pragmatic engineering approximation. The latent variables represent aggregated effects of multiple nonlinear modeling errors and approximation uncertainties, and empirical evidence indicates that their distribution is approximately unimodal and symmetric. Estimating more complex, parameter-dependent covariance structures is not statistically reliable given the limited HF dataset.

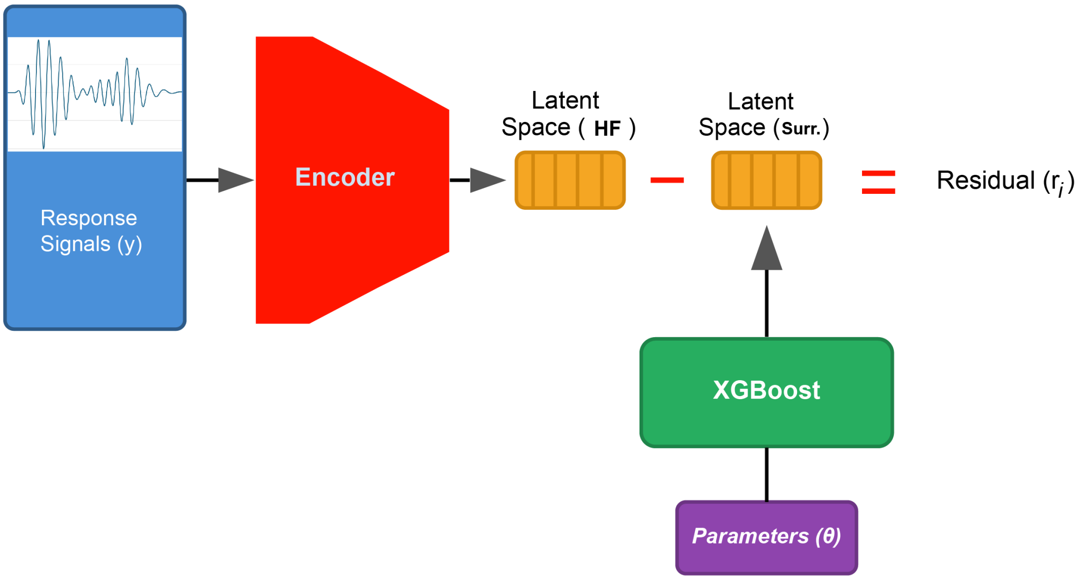


Figure 3.8: Computation of latent-space residuals for uncertainty quantification. The residual $\mathbf{r}_i$ is computed as the difference between the XGBoost-predicted latent code $\boldsymbol{\mu}_z(\boldsymbol{\theta}_i)$ and the encoder-derived latent representation $\mathrm{Enc}(\mathbf{u}_i^{\mathrm{HF}})$ for each high-fidelity sample.

3.7.1 Latent-space covariance estimation from residuals

The latent-space covariance matrix $\boldsymbol{\Sigma}_z$ quantifies the surrogate-induced uncertainty by capturing the discrepancy between predicted and latent representations for HF data by encoder. For each of the N_{HF} available HF samples $\{(\boldsymbol{\theta}_i, \mathbf{u}_i^{\mathrm{HF}})\}_{i=1}^{N_{\mathrm{HF}}}$, where $\boldsymbol{\theta}_i$ denotes the damage parameters and $\mathbf{u}_i^{\mathrm{HF}}$ is the corresponding HF response, a latent residual is computed as

$$\mathbf{r}_i = \boldsymbol{\mu}_z(\boldsymbol{\theta}_i) - \mathrm{Enc}(\mathbf{u}_i^{\mathrm{HF}}), \qquad i = 1, \ldots, N_{\mathrm{HF}}. \tag{3.14}$$

Here, $\boldsymbol{\mu}_z(\boldsymbol{\theta}_i)$ is the latent representation predicted by the fine-tuned XGBoost surrogate given the damage parameters, and $\mathrm{Enc}(\mathbf{u}_i^{\mathrm{HF}})$ is the latent representation obtained by encoding the HF response through the fine-tuned encoder. The residual $\mathbf{r}_i$ thus represents the error in the latent space predictions when applied to HF data. The computation of these residuals is illustrated in Figure 3.8.

An unbiased estimator of the latent covariance matrix is then computed as

$$\boldsymbol{\Sigma}_z = \frac{1}{N_{\mathrm{HF}} - 1} \sum_{i=1}^{N_{\mathrm{HF}}} (\mathbf{r}_i - \bar{\mathbf{r}})(\mathbf{r}_i - \bar{\mathbf{r}})^\top, \tag{3.15}$$

which measures the variance and covariance structure of the residuals across the latent space dimensions; here $\bar{\mathbf{r}}$ is the mean of residuals $\mathbf{r}_i$. This constant-covariance approximation assumes that the surrogate uncertainty is stationary across the damage parameter space and provides a stable, computationally efficient representation of the uncertainty quantification given the limited HF data available.

3.8 Propagating Uncertainty from Latent to Response Space

The decoder is a nonlinear mapping, and therefore the induced distribution of the decoded response $\mathbf{u}$ is generally non-Gaussian even when the latent distribution is Gaussian. To account for this nonlinearity and propagate the latent-space uncertainty to the response space, Monte Carlo sampling is employed. For a given parameter vector $\boldsymbol{\theta}$, K latent samples are drawn from the predicted latent distribution $\mathbf{z}^{(k)} \sim \mathcal{N}(\boldsymbol{\mu}_z(\boldsymbol{\theta}), \boldsymbol{\Sigma}_z)$, where $\boldsymbol{\mu}_z(\boldsymbol{\theta})$ is the XGBoost prediction and $\boldsymbol{\Sigma}_z$ is the estimated latent covariance. Each latent sample is then decoded through the fine-tuned decoder to obtain response samples $\mathbf{u}^{(k)} = \mathrm{Dec}(\mathbf{z}^{(k)})$. The uncertainty propagation process is illustrated in Figure 3.9.

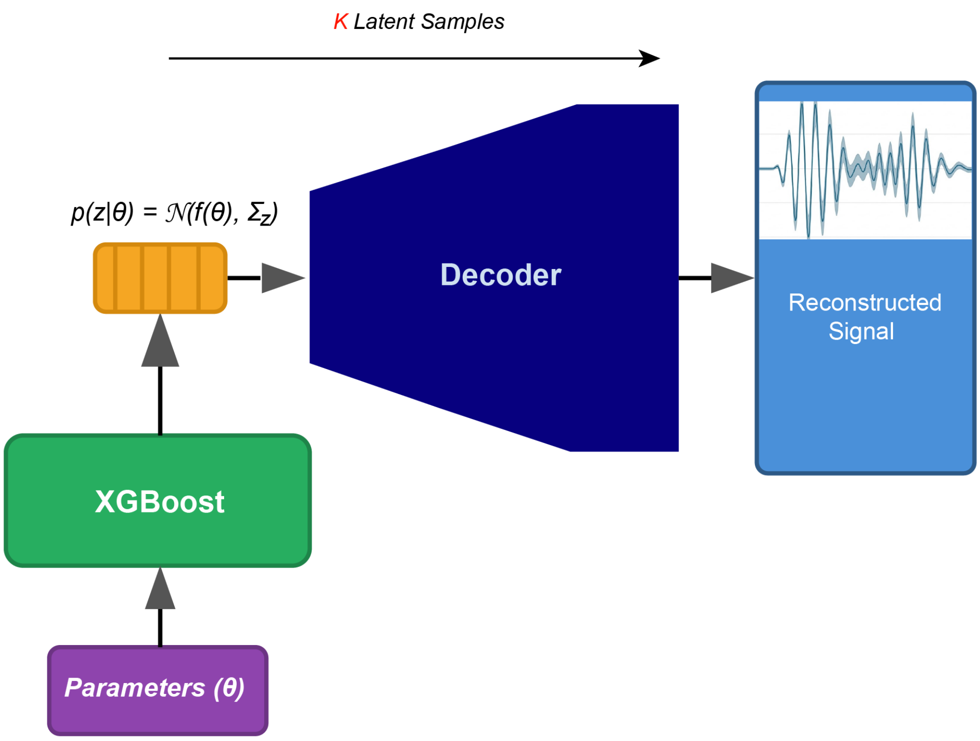


Figure 3.9: Uncertainty propagation through the decoder network. Monte Carlo samples drawn from the Gaussian latent distribution are decoded to obtain an ensemble of response predictions, from which the predictive mean and pointwise variance are estimated.

From the ensemble of decoded responses, the predictive mean and covariance are estimated as

$$\hat{\boldsymbol{\mu}}_u(\boldsymbol{\theta}) = \frac{1}{K} \sum_{k=1}^{K} \mathbf{u}^{(k)}, \tag{3.16}$$

$$\boldsymbol{\Sigma}_{\text{sur}}(\boldsymbol{\theta}) = \frac{1}{K-1} \sum_{k=1}^{K} (\mathbf{u}^{(k)} - \hat{\boldsymbol{\mu}}_u)(\mathbf{u}^{(k)} - \hat{\boldsymbol{\mu}}_u)^\top. \tag{3.17}$$

The response vector $\mathbf{u}$ has dimension 1500, which would result in a covariance matrix of size 1500×1500. For computational convenience and tractability in Bayesian inversion, only the diagonal of $\boldsymbol{\Sigma}_{\text{sur}}$ is retained. The diagonal matrix captures pointwise uncertainty at each response location while avoiding the significant computational overhead associated with full covariance matrices during likelihood evaluations and matrix operations. The mean and diagonal covariance matrix of the responses are used in defining an assumed Gaussian distributed likelihood, $p(\mathbf{u} \mid \boldsymbol{\theta}) \approx \mathcal{N}(\mathbf{u}; \hat{\boldsymbol{\mu}}_u(\boldsymbol{\theta}), \boldsymbol{\Sigma}_{\text{sur}}(\boldsymbol{\theta}))$ for Bayesian inference

Chapter 4

Bayesian Inverse Inference Using Multi-Fidelity Surrogate Model

Chapter 3 developed a probabilistic multifidelity surrogate model (MFSM) that predicts lamb wave responses with uncertainty estimates. This chapter addresses the inverse problem: given a measured wave response at the observation point, what can be inferred about the notch parameters that produced it using the MFSM? The three unknowns are notch location, depth, and width. The MFSM enables rapid evaluation of candidate parameter sets during inference, replacing expensive finite element simulations.

4.1 Inverse Problem Formulation

The forward problem maps damage parameters $\boldsymbol{\theta} = [x_d,\ w_d,\ d_d]^\top$ to the 1500-point wave response $\mathbf{u}(\boldsymbol{\theta})$ measured at receiver location $x = 1.9$ m. Deterministic inversion seeks a single best-fit solution by minimizing response mismatch, but this provides no measure of uncertainty or the possibility of multiple solutions, which is not unusual of inverse problems. When surrogate error is present, or multiple parameter combinations produce similar responses, point estimates can be misleading.

Bayesian inference treats the inverse problem probabilistically. Bayes' theorem expresses the posterior distribution over damage parameters as

$$p(\boldsymbol{\theta} \mid \mathbf{u}^{\text{obs}}) = \frac{p(\mathbf{u}^{\text{obs}} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})}{p(\mathbf{u}^{\text{obs}})} \tag{4.1}$$

where $p(\boldsymbol{\theta})$ encodes prior knowledge, $p(\mathbf{u}^{\text{obs}} \mid \boldsymbol{\theta})$ is the likelihood quantifying observation probability given parameters, and the denominator normalizes the distribution. The observation of the response may also incorporate measurement noise:

$$\mathbf{u}^{\text{obs}} = \mathbf{u} + \boldsymbol{\epsilon}; \quad \boldsymbol{\epsilon} \overset{i.i.d}{\sim} \mathcal{N}(\mathbf{0}, \gamma^2 \mathbf{I}) \tag{4.2}$$

The posterior captures everything the data reveal about the damage. A narrow distribution indicates well-determined parameters. Multiple peaks indicate competing damage configurations equally consistent with measurements.

For the notch characterization problem, uniform priors are assigned over physically feasible ranges:

$$p(\boldsymbol{\theta}) = \begin{cases} \dfrac{1}{\text{Vol}(\Theta)} & \text{if } x_d \in [1.6545, 1.8355] \text{ m}, w_d \in [0.1, 1.2] \text{ mm}, d_d \in [0.1, 1.0] \text{ mm} \\ 0 & \text{otherwise} \end{cases} \tag{4.3}$$

These bounds reflect the training domain of the surrogate. Depths and widths are transformed to a logarithmic scale during inference to improve sampling efficiency across orders of magnitude.

4.2 Likelihood Function with Surrogate Uncertainty

The likelihood $p(\mathbf{u}^{\text{obs}} \mid \boldsymbol{\theta})$ quantifies how probable the observed time-series response is given candidate damage parameters. The MFSM from Chapter 3 predicts not a single response but a distribution where $\mathbf{u}(\boldsymbol{\theta}) \sim \mathcal{N}\left(\hat{\boldsymbol{\mu}}_u(\boldsymbol{\theta}), \boldsymbol{\Sigma}_{\text{sur}}\right)$, with $\hat{\boldsymbol{\mu}}_u(\boldsymbol{\theta})$ as the mean prediction and $\boldsymbol{\Sigma}_{\text{sur}}$ capturing surrogate model uncertainty arising from the autoencoder-XGBoost approximation and finite training data.

An assumed Gaussian likelihood incorporating this uncertainty takes the form

$$p(\mathbf{u}^{\text{obs}} \mid \boldsymbol{\theta}) = \mathcal{N}\left(\mathbf{u}^{\text{obs}}; \hat{\boldsymbol{\mu}}_u(\boldsymbol{\theta}), \boldsymbol{\Sigma}_{\text{sur}} + \gamma^2 \mathbf{I}\right). \tag{4.4}$$

As discussed in Section 3.8, the covariance matrix is retained in diagonal form to capture pointwise uncertainty at each response component

$$\boldsymbol{\Sigma}_{\text{sur}} = \text{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_{n_t}^2), \tag{4.5}$$

where each σ_i^2 represents the surrogate-induced uncertainty at the i-th response location. With this diagonal structure, the log-likelihood simplifies to

$$\log p(\mathbf{u}^{\text{obs}} \mid \boldsymbol{\theta}) = -\frac{1}{2} \sum_{t=1}^{n_t} \frac{(u_t^{\text{obs}} - \hat{\mu}_{u,t}(\boldsymbol{\theta}))^2}{\sigma_t^2 + \gamma^2} - \sum_{t=1}^{n_t} \log(\sigma_t + \gamma^2) - \frac{n_t}{2} \log(2\pi). \tag{4.6}$$

The pointwise variances σ_t^2 are precomputed during surrogate training. For each of the N_{HF} parameter vectors, K latent samples are drawn from the predicted latent distribution, decoded through the fine-tuned autoencoder, and the empirical variance at each response index is computed across the decoded ensemble.

4.3 Differential Evolution-Accelerated Parallel Tempering MCMC

Computing the posterior analytically is intractable for this nonlinear problem. Markov Chain Monte Carlo generates samples from the posterior by constructing a chain with the correct stationary distribution. The Metropolis-Hastings algorithm begins at an initial state $\boldsymbol{\theta}^{(i)}$ and proposes a new state $\boldsymbol{\theta}^*$ from a proposal distribution. For symmetric proposals like random walks, the acceptance probability is

$$\alpha = \min\left(1, \frac{p(\mathbf{u}^{\mathrm{obs}} \mid \boldsymbol{\theta}^*)\, p(\boldsymbol{\theta}^*)}{p(\mathbf{u}^{\mathrm{obs}} \mid \boldsymbol{\theta}^{(i)})\, p(\boldsymbol{\theta}^{(i)})}\right) \tag{4.7}$$

A random walk proposal perturbs the current state by Gaussian noise:

$$\boldsymbol{\theta}^* = \boldsymbol{\theta}^{(i)} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\mathrm{prop}}) \tag{4.8}$$

Proposal step sizes control exploration versus exploitation. Small steps yield high acceptance but slow convergence; large steps cause frequent rejection. Adaptive tuning during sampling adjusts $\boldsymbol{\Sigma}_{\mathrm{prop}}$ to achieve acceptance rates around 25%.

4.3.1 Smart initialization using differential evolution

Standard MCMC from random starting points wastes iterations traversing low-probability regions before finding the posterior. Differential Evolution locates high-probability regions first. A population of N_p candidates evolves through mutation and selection. For each candidate $\mathbf{x}_i$, a mutant is formed:

$$\mathbf{v}_i = \mathbf{x}_{r_1} + F \cdot (\mathbf{x}_{r_2} - \mathbf{x}_{r_3}) \tag{4.9}$$

where r_1, r_2, r_3 are distinct random indices and $F \approx 0.8$ controls perturbation scale. Selection keeps the better of the original and perturbed candidates. After 100 generations with a population size of 80, the best member initializes the MCMC chain. This strategy dramatically reduces burn-in and reduces computational time.

4.3.2 MCMC implementation details

The inverse inference procedure estimates posterior distributions over notch location x_d, depth d_d, and width w_d given an observed 1500-point wave response. The likelihood from Equation 4.6 quantifies how well the surrogate prediction at candidate parameters matches the observed response, weighted by the surrogate uncertainty at each response point. Parameters producing predictions close to the observation receive high likelihood, while those producing large residuals relative to the surrogate variance receive low likelihood.

The inference begins with Differential Evolution to locate a high-quality starting point. A population of 80 candidate parameter vectors evolves over 30 generations through mutation and selection. After 30 generations, the best population member provides an initial estimate lying in a high-probability region of the posterior.

From this starting point, 48 parallel Markov chains are initialized for parallel tempering. Of these, 29 chains start near the DE solution with small Gaussian perturbations, while the remaining 19 chains are dispersed uniformly across the prior domain. This mixed initialization strategy balances exploitation of the known good region with exploration that might discover alternative modes.

The 48 chains are organized into 6 temperature levels with 8 chains per level. Temperatures follow a geometric schedule from $T = 1$ (cold) to $T = 20$ (hot). Temperature modifies the likelihood contribution to the acceptance probability: at temperature T, the tempered log-posterior becomes

$$\log p_T(\boldsymbol{\theta} \mid \mathbf{u}^{\text{obs}}) \propto \frac{1}{T} \log p(\mathbf{u}^{\text{obs}} \mid \boldsymbol{\theta}) + \log p(\boldsymbol{\theta}) \tag{4.10}$$

Cold chains at $T = 1$ sample from the true posterior. Hot chains sample from flattened distributions where the likelihood surface is smoothed, allowing them to traverse between modes that cold chains cannot cross. Every 10 iterations, adjacent temperature levels attempt replica exchange: two chains swap their parameter states with probability depending on their respective likelihoods and temperatures. This mechanism allows favorable configurations discovered by hot chains to propagate down to cold chains, improving mixing and mode discovery.

Each chain runs for 5000 iterations using Metropolis-Hastings updates. A random walk proposal perturbs the current state where proposal scales adapt during sampling to maintain acceptance rates near 25%. Notch depth and width are sampled on logarithmic scale to improve mixing across their order-of-magnitude ranges.

After sampling completes, the first 10% of iterations are discarded as burn-in, leaving 4000 iterations per chain for posterior analysis. Only samples from cold chains ($T = 1$) represent the true posterior; samples from heated chains are used solely to improve mixing and are discarded. With 8 cold chains, the final posterior is characterized by $8 \times 4000 = 32000$ samples.

The maximum *a posteriori* (MAP) estimate is identified as the sample with highest posterior probability. For multimodal posteriors, kernel density estimation reveals distinct peaks, and each mode is characterized separately by its location and local credible intervals. The posterior samples provide uncertainty quantification through credible intervals containing specified probability mass.

Chapter 5

Results and Discussion

This chapter presents the results obtained from the proposed multi-fidelity surrogate modeling framework and its application to inverse damage characterization. The discussion is organized into two parts: the first addresses the performance of the forward surrogate model, while the second examines the Bayesian inverse inference results. Before presenting these findings, the evaluation metrics and preliminary validation studies are described.

5.1 Performance Metrics and Validation Approach

Three metrics are used to assess the predictive accuracy of the surrogate models: the coefficient of determination (R^2), mean squared error (MSE), and normalized mean squared error (NMSE).

The coefficient of determination (R^2) quantifies the proportion of variance in the target response that is explained by the model. For a set of n test samples with true responses y_i and predicted responses $\hat{y}_i$, it is computed as

$$R^2 = 1 - \frac{\sum_{t=1}^{n_t} (y_t - \hat{y}_t)^2}{\sum_{t=1}^{n_t} (y_t - \bar{y})^2} \tag{5.1}$$

where $\bar{y}$ denotes the mean of the true responses. A value of $R^2 = 1$ indicates agreement between predictions and targets, while values approaching zero or negative indicate that the model fails to capture the underlying variance.

The mean squared error measures the average squared deviation between predictions and targets:

$$\text{MSE} = \frac{1}{n_t} \sum_{t=1}^{n_t} (y_t - \hat{y}_t)^2 \tag{5.2}$$

This metric is scale-dependent and provides an absolute measure of prediction error in the units of the response variable squared.

The normalized mean squared error expresses the MSE as a percentage of the total variance in the target data:

$$\text{NMSE} = \frac{\text{MSE}}{\text{Var}(y(t))} \times 100\% \tag{5.3}$$

where $\text{Var}(y(t))$ is the variance of the true response signal. NMSE facilitates comparison across datasets with different scales and provides an intuitive interpretation of model performance relative to the signal variability.

5.2 High-Fidelity Model Verification

Before training the multi-fidelity surrogate, the in-house two-dimensional finite element model was validated against ANSYS Mechanical. Same 8 noded element and same number of elements in both length and depth direction were used to perform this comparison test. The comparison was performed for a notched aluminum beam with the notch located at $x = 1.75$ m from the left boundary. The notch dimensions were set to the largest values in the parametric range: depth of 1 mm and width of 1.2 mm. These parameters were chosen because larger notches produce stronger wave scattering, making any discrepancies between the two solvers more apparent.

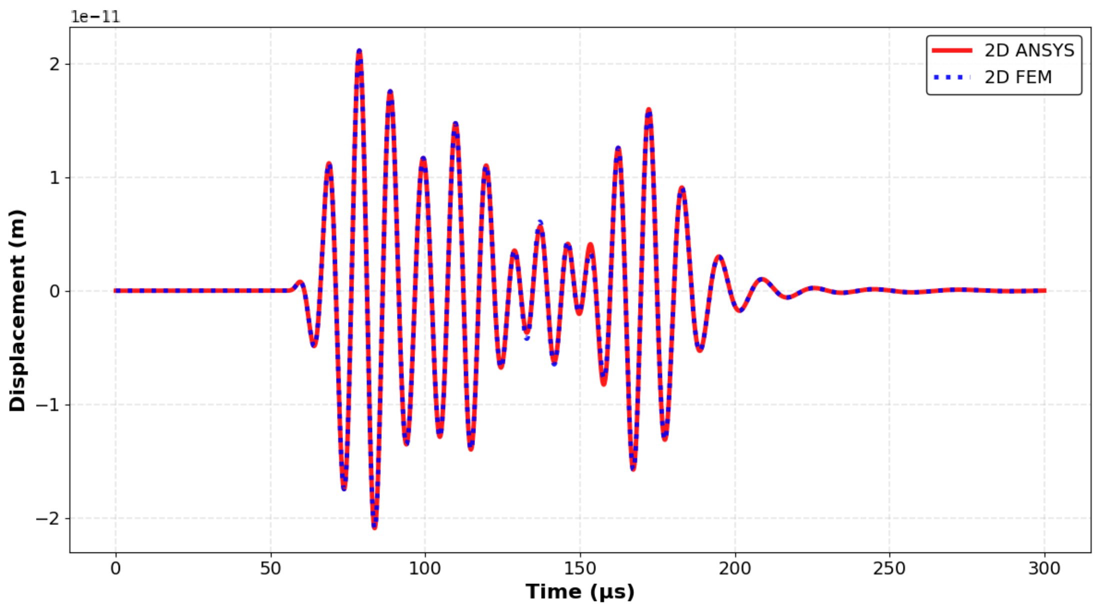


Figure 5.1: Validation of the in-house 2D FEM solver against ANSYS Mechanical for a notched beam ($x_d = 1.75$ m, $d_d = 1$ mm, $w_d = 1.2$ mm). The two solutions achieve $R^2 \approx 0.99$ agreement in the axial displacement response at the sensing location.

Figure 5.1 shows the axial displacement time history at the sensing location obtained from both solvers. The in-house model achieves approximately 99%

agreement with the ANSYS solution, as measured by R^2. Minor differences are observed in the wave packets generated by notch interaction. These discrepancies may be attributed to differences in element formulation between the two solvers. The in-house implementation uses Q8 quadrilateral elements with full integration, while ANSYS employs higher-order interpolation and selective reduced integration schemes by default. Additionally, the treatment of notch boundaries differs between the two implementations. The in-house model represents the notch through mesh refinement at notch region, whereas ANSYS applies mesh smoothing algorithms that may alter the local stress concentration behavior. Despite these minor variations, the overall agreement validates the use of the in-house solver for generating high-fidelity training data.

5.3 Autoencoder Architecture Selection

The autoencoder component of the surrogate model was designed to compress the 1500-dimensional response signal into a 30-dimensional latent representation. Two architectures were considered: a convolutional neural network (CNN) based encoder-decoder and a multilayer perceptron (MLP) based architecture.

Preliminary experiments revealed that the CNN-based autoencoder exhibited greater variability in reconstruction accuracy across the test set compared to the MLP architecture. Distributions of R^2 scores for predictions obtained using a CNN architecture and an MLP architecture are shown in Figure 5.2, and it can be seen that the training errors were far more (long tails) for CNN than for MLP. Although CNNs are often preferred for sequential data due to their ability to capture local patterns through convolutional filters, the Lamb wave responses in this study favored an MLP-based autoencoder architecture instead. The response signals contain both localized features, such as the scattered wave packets from notch interaction, and global features, including the fundamental wavefronts that propagate across the entire time window. The MLP architecture, with its fully connected layers, captures dependencies across the full temporal extent of the signal.

5.4 Forward Surrogate Model Performance

Three surrogate configurations were evaluated on a test set comprising 50 high-fidelity simulations that were not used during training. The multi-fidelity surrogate model (MFSM) was trained using the two-phase approach described in Chapter 3, with 750 low-fidelity samples for initial autoencoder and XGBoost training and 100 high-fidelity samples for fine-tuning. Two baseline models were also constructed for comparison: a high-fidelity-only surrogate model (HFSM) trained exclusively on the 100 high-fidelity samples, and a low-fidelity-only surrogate model (LFSM) trained exclusively on the 750 low-fidelity samples without any fine-tuning.

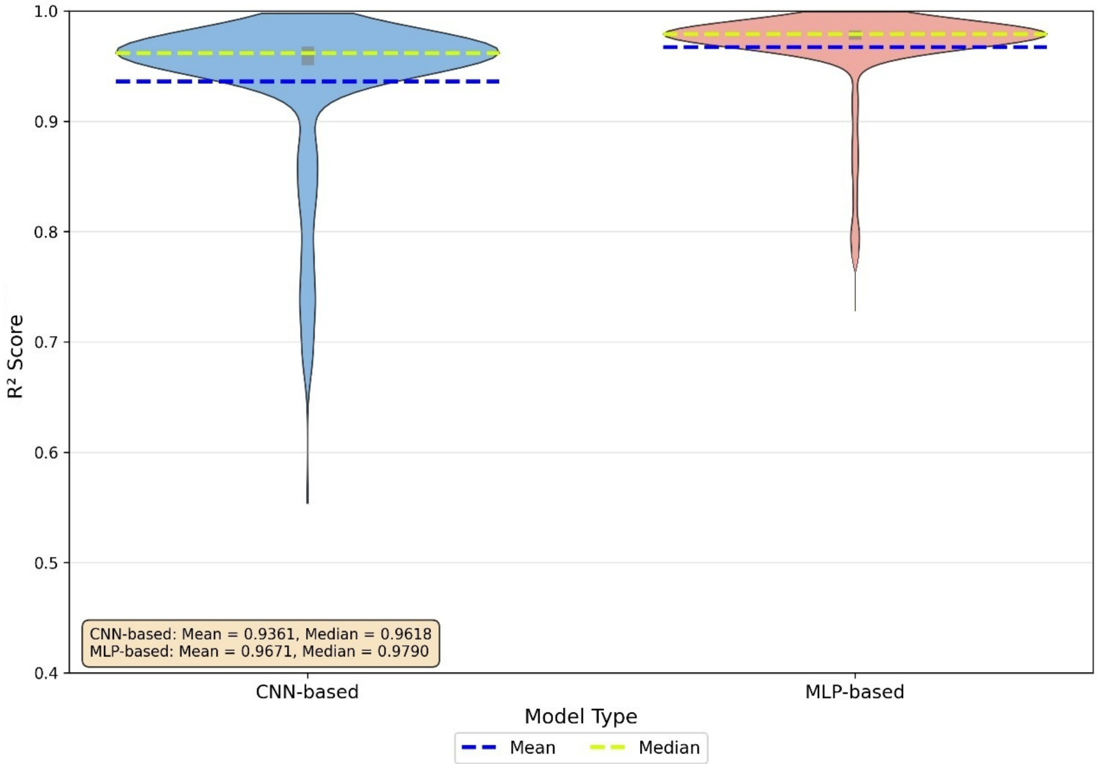


Figure 5.2: Distribution of R^2 reconstruction accuracy across test cases for CNN-based and MLP-based autoencoder architectures. The MLP architecture exhibits lower variance and fewer outliers in the lower tail.

Table 5.1: Performance comparison across surrogate models on 50 high-fidelity test cases

Model	$\mathbf{R^2}$ Score	**MSE**	**NMSE**
MFSM (Multi-Fidelity)	0.9693	0.00159	2.66%
HFSM (High-Fidelity)	0.8703	0.00660	11.15%
LFSM (Low-Fidelity)	0.8324	0.00905	16.74%

Table 5.1 summarizes the performance metrics for each surrogate configuration. The multi-fidelity surrogate achieves an R^2 of 0.9693, corresponding to an NMSE of 2.66%. The high-fidelity-only model attains an R^2 of 0.8703 with an NMSE of 9.15%, while the low-fidelity-only model yields an R^2 of 0.8324 and an NMSE of 16.74%.

The results demonstrate that the multi-fidelity framework outperforms both single-fidelity baselines. The HFSM, despite being trained on the same 100 high-fidelity samples used for fine-tuning, achieves lower accuracy than the MFSM. This indicates that the initial training on low-fidelity data provides a beneficial initialization for the autoencoder, enabling it to learn fundamental wave propagation patterns before fine-tuning on the limited high-fidelity data. The LFSM performs worst among the three configurations, which is expected since it was trained exclusively on zigzag theory simulations that systematically differ from the 2D finite element responses. The following violin plot 5.3 shows the variation of R^2 in MFSM and HFSM test set prediction. It clearly shows that MFSM gives consistent performance with less variation in accuracy over the test set.

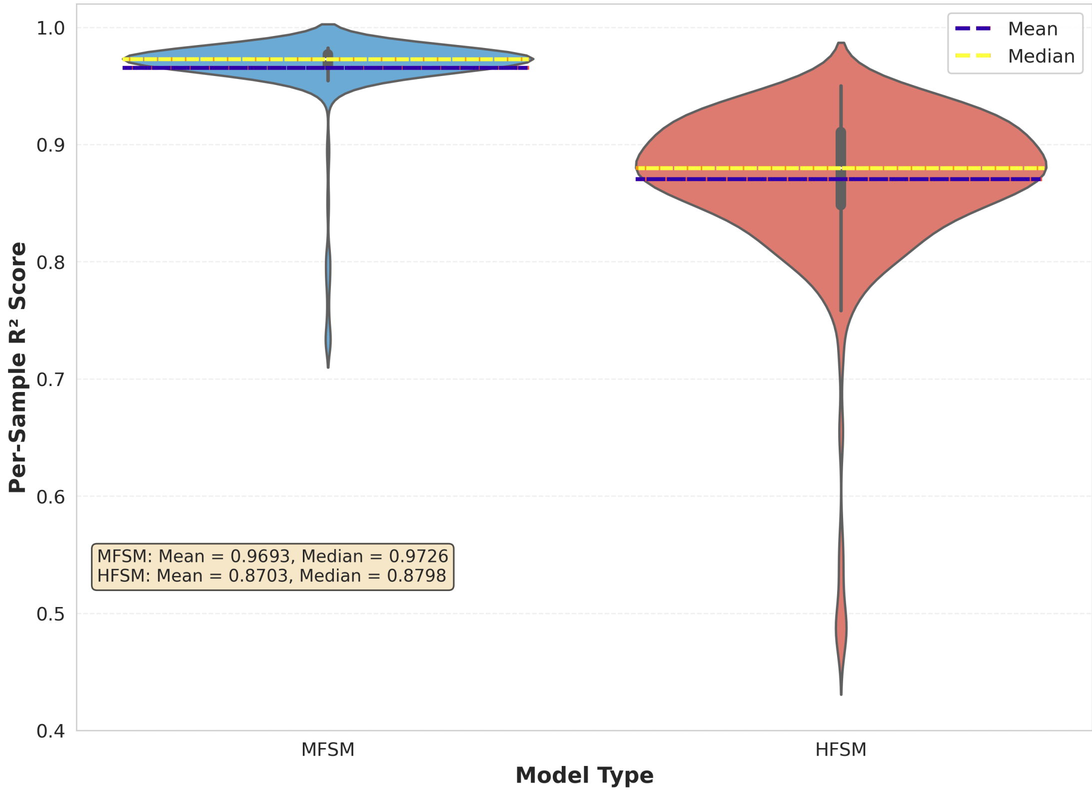
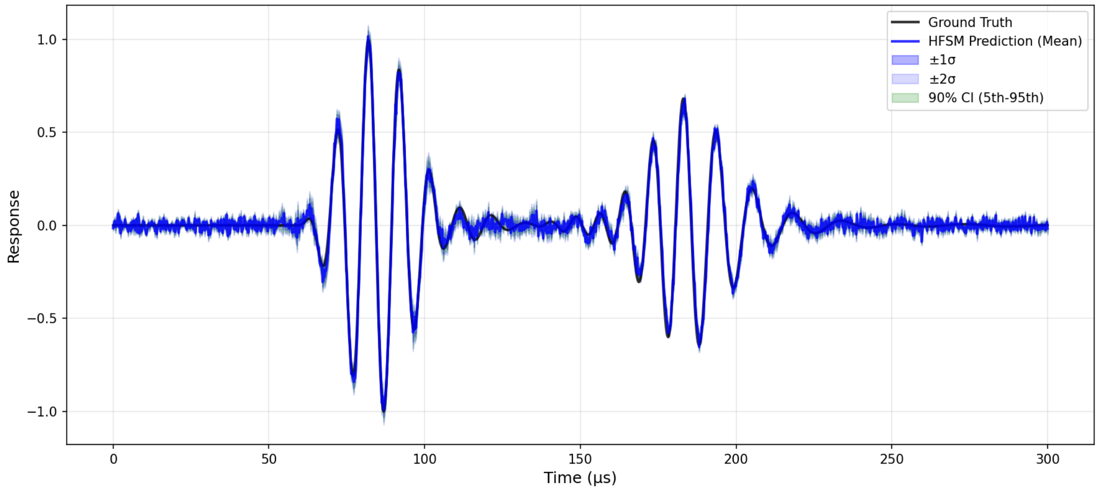


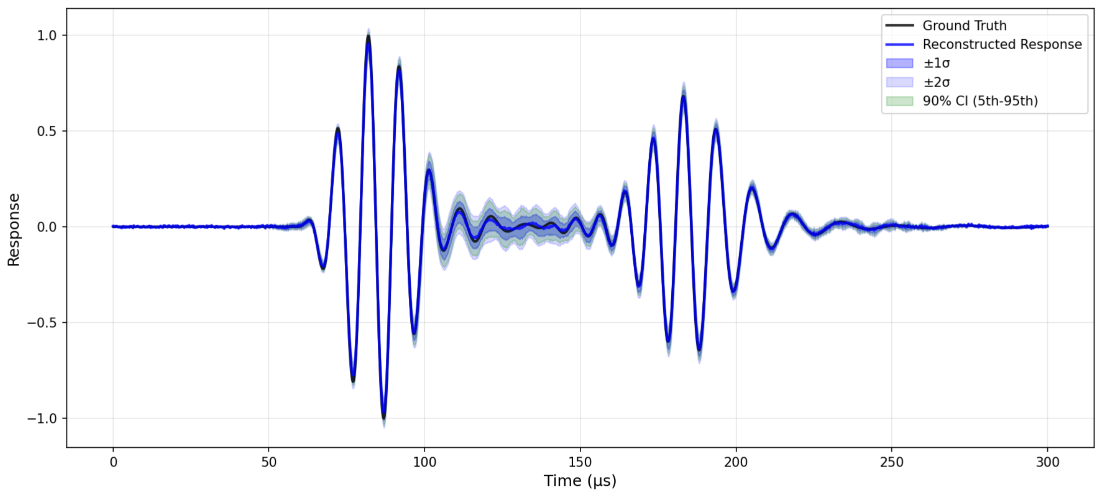
Figure 5.3: Violin plot comparing R^2 distributions on the 50-sample high-fidelity test set for the multi-fidelity surrogate model (MFSM) and high-fidelity surrogate model (HFSM). The MFSM achieves higher median accuracy with substantially reduced variance.

5.4.1 Best-case reconstruction performance

Figure 5.4 presents the best-case prediction from the HFSM alongside the MFSM prediction for the same damage configuration. The HFSM achieves an R^2 score of 0.981 but exhibits noticeable noise fluctuations in the quiescent regions and particularly in the middle portion where notch-induced scattering occurs. Despite these imperfections, it successfully captures the arrival times and overall structure of both symmetric and anti-symmetric wavemodes. In contrast, the MFSM prediction, with an R^2 score of 0.998, demonstrates superior fidelity by accurately reproducing the complete wave physics including the subtle secondary wave packets generated by notch interaction. The enhanced performance stems from the additional low-fidelity training data that enables the autoencoder to develop a more robust latent representation capable of capturing fine structural details critical for precise damage parameter estimation.



(a) HFSM best prediction ($R^2 = 0.981$)

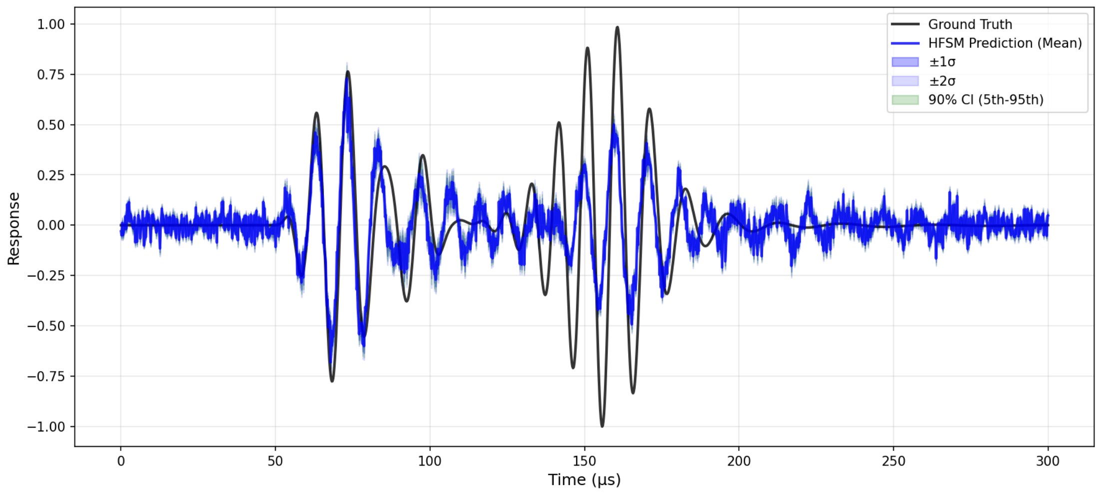


(b) MFSM prediction for the same case ($R^2 = 0.998$)

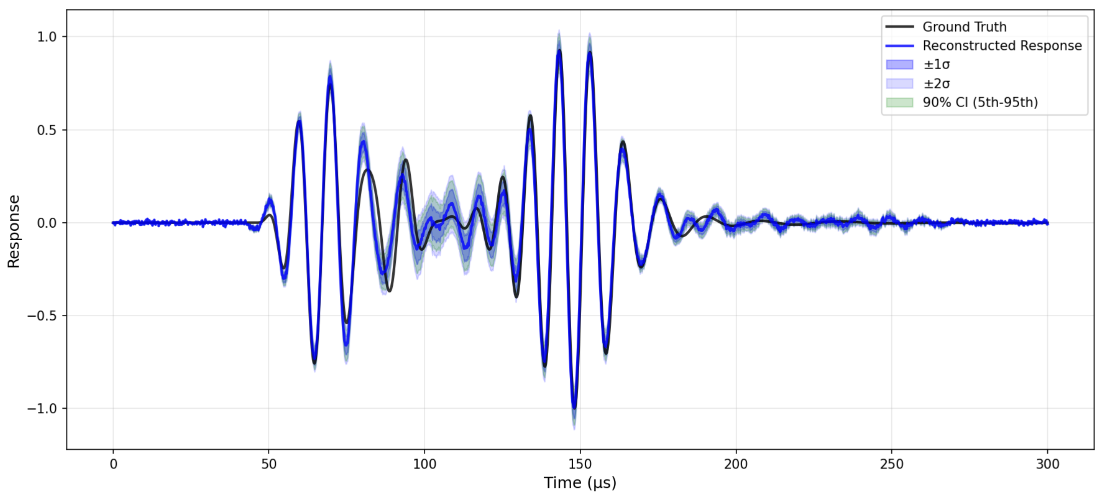
Figure 5.4: Response predictions for the best-performing HFSM test case ($x_d = 1.7321$ m, $d_d = 0.3$ mm, $w_d = 0.8$ mm): (a) HFSM prediction with $R^2 = 0.981$, (b) MFSM prediction for the same configuration with $R^2 = 0.998$. Shaded regions indicate $\pm 2\sigma$ uncertainty bounds.

5.4.2 Worst-case reconstruction performance

Figure 5.5 shows the worst-case prediction from the HFSM compared with the MFSM prediction for the same damage configuration representing a deep and wide notch. The HFSM prediction, with an R^2 score of 0.465, fails catastrophically in capturing the fundamental wave physics. The model produces incorrect wave-mode shapes for both symmetric and anti-symmetric components, exhibits severe amplitude errors, and introduces substantial noise throughout the response. This prediction would be essentially useless for any practical damage characterization task.



(a) HFSM worst prediction ($R^2 = 0.465$)



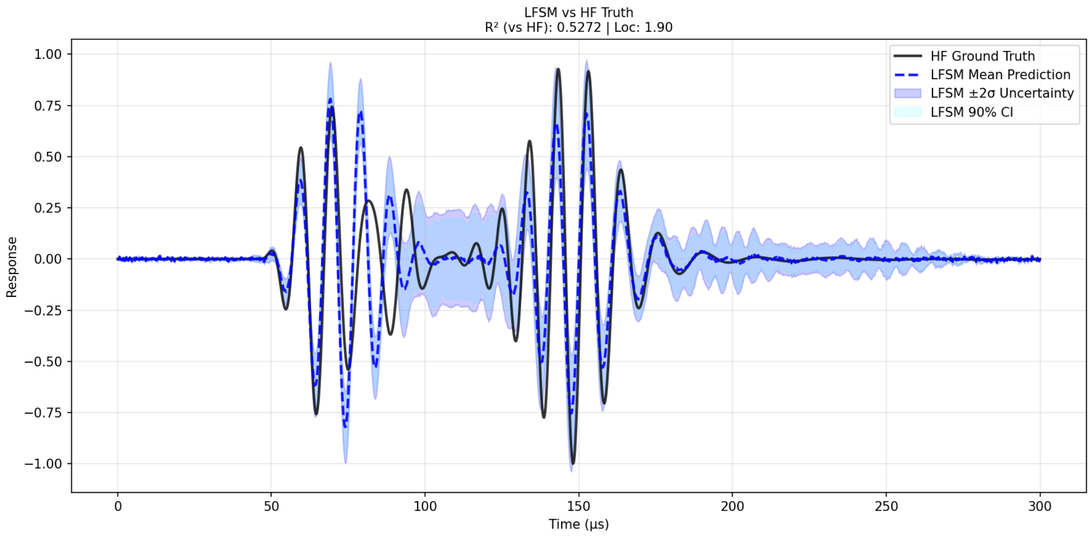
(b) MFSM prediction for the same case ($R^2 = 0.928$)

Figure 5.5: Response predictions for the worst-performing HFSM test case ($x_d = 1.6804$ m, $d_d = 1.0$ mm, $w_d = 1.2$ mm): (a) HFSM prediction with $R^2 = 0.465$ showing substantial waveform distortion, (b) MFSM prediction with $R^2 = 0.928$ maintaining reasonable fidelity. This case represents a deep, wide notch near the parameter domain boundary.

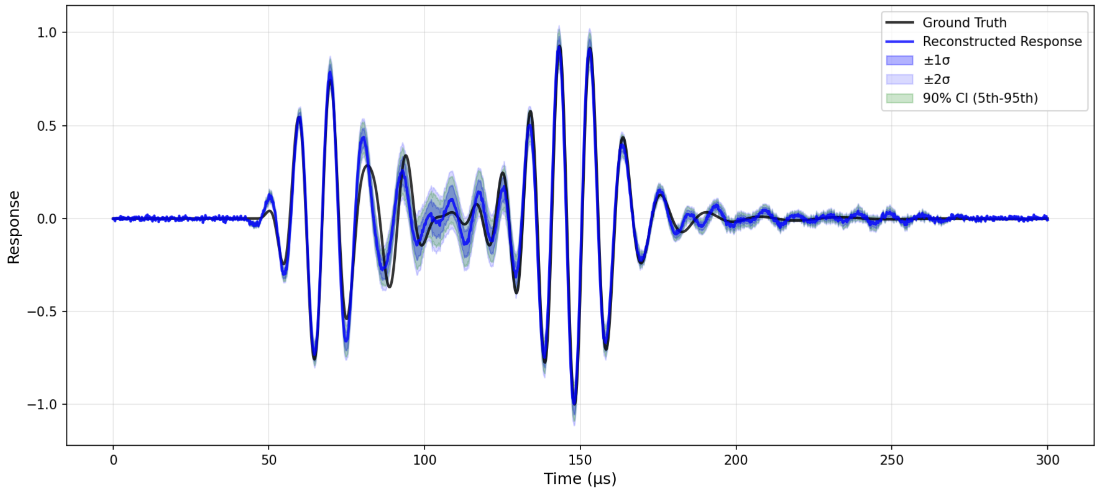
The MFSM prediction for the identical case achieves an R^2 score of 0.928 and, while showing some deterioration compared to the best case, maintains reasonable fidelity to the target response. The model preserves the correct wavemode shapes and timing, and notably succeeds in approximately capturing the complex interaction effects in the middle portion where the scattered waves interfere with the primary wavemodes. This comparison demonstrates the MFSM's significantly improved robustness when dealing with challenging damage configurations near the boundaries of the training domain.

5.4.3 Low-fidelity versus multi-fidelity surrogate comparison

Figure 5.6 compares the performance of the low-fidelity surrogate model against the multi-fidelity surrogate model, demonstrating the value added by the MFSM framework. The LFSM prediction, when evaluated against high-fidelity ground truth, achieves an R^2 score of only 0.527 and exhibits fundamental deficiencies in capturing the wave physics. The model fails to accurately represent the shape evolution of the symmetric wavemode and significantly underpredicts the amplitude of the anti-symmetric wavemode. Additionally, the uncertainty bands are substantially wider than those observed in the MFSM predictions. This increased uncertainty arises because the high-fidelity data is being passed through an encoder trained exclusively on low-fidelity responses, causing the latent space representation to differ markedly from the distribution on which the surrogate's covariance matrix and uncertainty quantification were based. The MFSM, by incorporating high-fidelity data into both the encoder training and the latent space regression, produces significantly tighter uncertainty bounds and more accurate predictions, validating the multi-fidelity approach.



(a) LFSM prediction compared against high-fidelity truth



(b) MFSM prediction for the same case

Figure 5.6: Performance comparison between low-fidelity-only (LFSM) and multi-fidelity (MFSM) surrogates evaluated against high-fidelity ground truth: (a) LFSM prediction ($R^2 = 0.527$) with wide uncertainty bounds, (b) MFSM prediction for the same case showing improved accuracy and tighter uncertainty estimates.

5.5 Surrogate Performance Sensitivity to Training Sample Size

The performance comparison between the MFSM and the HFSM reveals the fundamental advantage of leveraging multiple information sources in data-limited scenarios. Figure 5.7 presents the R^2 scores for both modeling approaches as a function of the number of high-fidelity training cases, spanning from 10 to 100 samples.

The HFSM, which learns directly from high-fidelity 2D FEM simulations without incorporating low-fidelity information, exhibits severe performance degradation when trained on sparse datasets. With only 10 high-fidelity samples, the HFSM achieves an R^2 score near zero, indicating essentially no predictive capability. The model shows rapid improvement as training data increases, reaching $R^2 = 0.72$ at 20 samples and gradually approaching $R^2 = 0.87$ at 100 samples. This behavior reflects the standard machine learning principle that complex mappings require substantial training data. The wave propagation problem involves mapping three damage parameters to 1500-point time-series responses through nonlinear physics, which presents a challenging learning task for models relying solely on limited high-fidelity data.

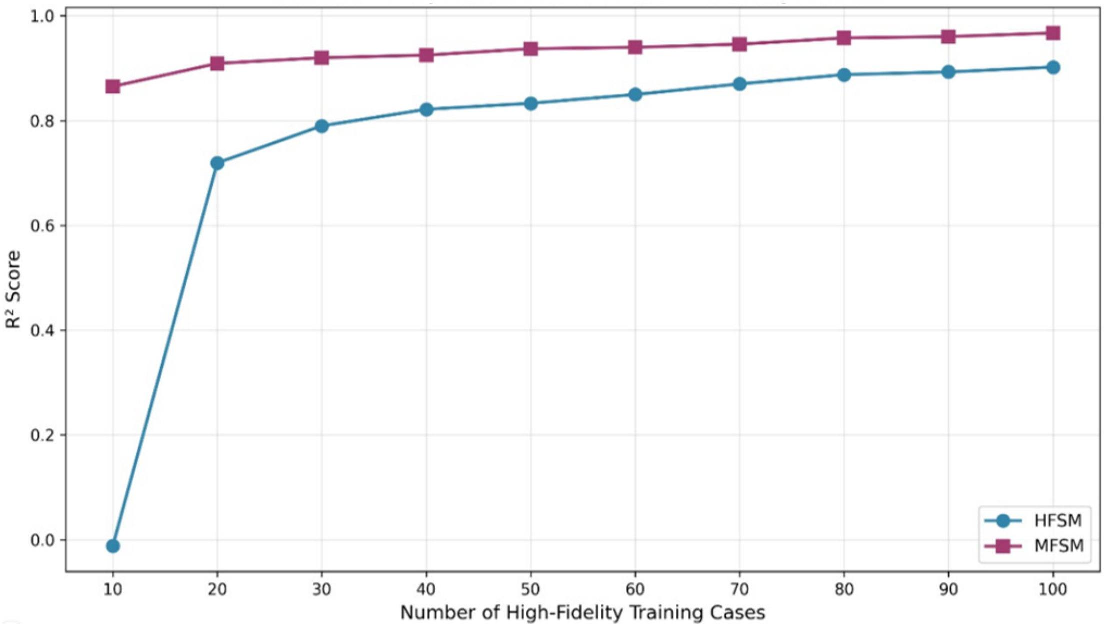


Figure 5.7: Test set R^2 as a function of high-fidelity training samples for MFSM and HFSM. The MFSM achieves $R^2 = 0.84$ with only 10 high-fidelity samples due to transfer learning from 750 low-fidelity simulations, while HFSM performance degrades severely below 50 samples.

In contrast, the MFSM demonstrates remarkable performance even with minimal high-fidelity training data. Starting at $R^2 = 0.84$ with only 10 high-fidelity samples, the MFSM maintains consistently superior accuracy across all sample sizes evaluated. The performance curve rises steadily but gradually, reaching $R^2 = 0.97$ at 100 samples. This sustained high performance stems from the transfer learning architecture employed in the MFSM framework. The model leverages 750 low-fidelity samples from the Refined Zigzag Theory to pre-train the autoencoder, establishing a robust latent representation of wave propagation patterns. The subsequent fine-tuning phase with limited high-fidelity data then refines this representation to capture the accuracy of 2D FEM simulations without requiring extensive high-fidelity training samples.

The performance gap between MFSM and HFSM is most pronounced in the low-sample regime. At 10 high-fidelity samples, the MFSM outperforms the HFSM by approximately 0.87 R^2 points. This gap narrows as more high-fidelity data becomes available, reducing to roughly 0.07 R^2 points at 100 samples. However, the practical implications favor the MFSM throughout this range. Each high-fidelity simulation requires 30-40 minutes of computational time, making datasets beyond 100 samples economically and temporally costly. The MFSM achieves near-optimal performance ($R^2 = 0.93$) with just 50 high-fidelity samples, whereas the HFSM requires more than 100 samples to reach comparable accuracy.

This comparative analysis underscores a critical insight for surrogate modeling in computational mechanics. When high-fidelity simulations are expensive but lower-fidelity alternatives exist, multi-fidelity approaches provide substantial advantages over single-fidelity methods. The MFSM's ability to maintain $R^2 > 0.84$ with as few as 10 high-fidelity samples makes it viable for applications where extensive high-fidelity datasets are impractical. For the structural health monitoring problem addressed in this work, where real-time damage characterization requires fast surrogate models trained on limited computational budgets, the multi-fidelity framework offers a practical path forward.

5.6 Inverse Inference Results

The inverse problem was formulated as Bayesian parameter estimation, where the posterior distribution over notch parameters was sampled using Differential Evolution Parallel Tempering MCMC (DE-PT-MCMC). The forward surrogate model provides rapid response predictions during sampling, enabling the generation of thousands of posterior samples within a computationally feasible timeframe.

5.6.1 Global sensitivity analysis: parameter influence on response

Before examining inverse estimation results, variance-based sensitivity analysis was performed to quantify the relative importance of each notch parameter on the measured wave response. The methodology follows the framework presented in Section 2.6 , applying Sobol indices to characterize parameter influence across the complete response time series.

The analysis employed the multi-fidelity surrogate model (MFSM) as the forward model, enabling efficient computation of sensitivity indices that would be prohibitively expensive using direct finite element simulations. Sample matrices were generated using the Saltelli sampling scheme with $N = 10000$ base samples, requiring $N(d + 2) = 50000$ surrogate evaluations for the three notch parameters (location, depth, width). The surrogate's millisecond-scale evaluation time made this analysis computationally feasible, whereas equivalent analysis using 2D finite element models would have required approximately 31,000 hours of computation time.

Each surrogate evaluation produced a 1500-point time series representing the sensor response. Sobol indices were computed pointwise at each time sample using the estimators given in Equations 2.11 and 2.12. This pointwise calculation yielded time-varying sensitivity profiles revealing how parameter influence evolves as different wave packets arrive at the sensor location.

The pointwise indices were aggregated to obtain global measures of parameter importance across the entire response window. Figure 5.8 presents these aggregated first-order and total-effect Sobol indices. Notch location exhibits the highest sensitivity. Notch depth ranks second in importance, contributing approximately similarly in the total variance. Notch width shows minimal sensitivity, with indices below 0.1.

This ordering aligns with physical expectations from guided wave propagation. The notch location determines when scattered waves arrive at the sensor relative to the incident wave packets. Small changes in location produce measurable time shifts in the response that are readily detectable. The scattered wave amplitude depends on notch depth, with deeper notches creating stronger reflections and mode conversions. These amplitude variations contribute variance but less distinctly than timing shifts. Notch width, within the parameter ranges considered, produces subtler effects on the scattering pattern that do not substantially alter either timing or amplitude characteristics captured by the sensor.

The sensitivity analysis provides essential context for interpreting inverse estimation results. Parameters with high Sobol indices contain more information about the measured response and should be more reliably estimated from data. Parameters with low indices contribute little to response variance, meaning measurement data provide limited information for inferring those parameters. The posterior uncertainty for width estimation, encountered in subsequent inverse results, is anticipated by its low sensitivity. Conversely, the accurate estimation of location and depth is supported by their dominant contributions to response variability.

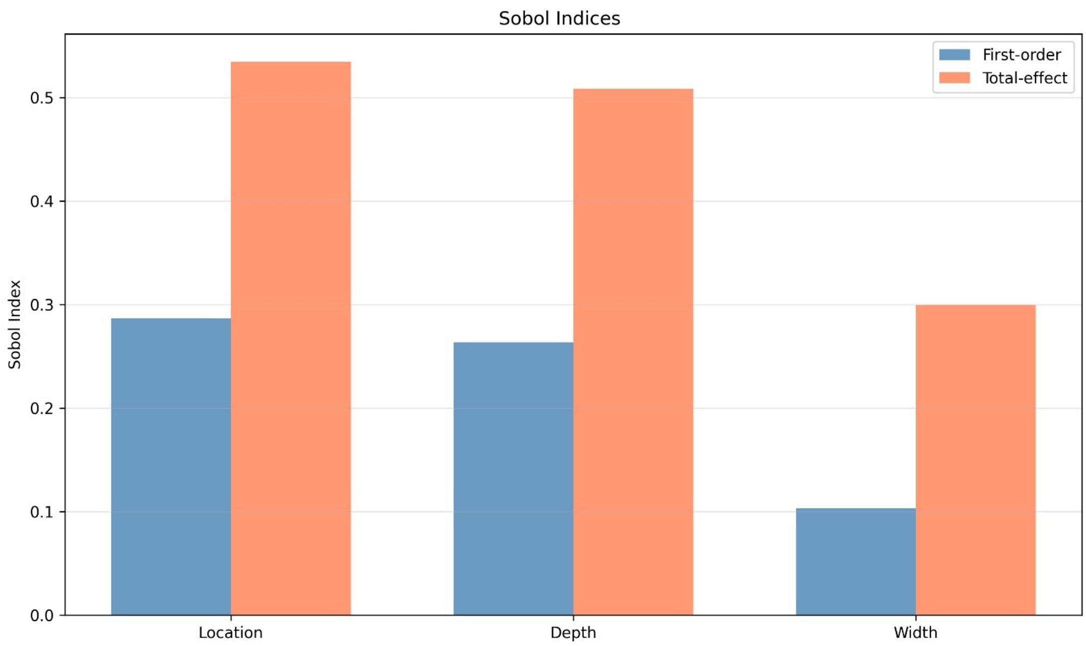


Figure 5.8: First-order (S_i) and total-effect (S_{T_i}) Sobol sensitivity indices for the three notch parameters, computed using $N = 10000$ base samples. Location and depth dominate response variance, while width exhibits minimal sensitivity.

5.6.2 Point estimates and error distributions

The inverse analysis was performed on several test cases using the MFSM as the forward model within the DE-PT-MCMC framework. For each test case, the maximum *a posteriori* (MAP) estimate was extracted from the posterior samples and compared against the true parameter values. Figure 5.9 presents scatter plots comparing estimated and true values for each parameter, along with histograms of the estimation errors. Location and depth estimates cluster around the diagonal line for a majority of the test cases, indicating reasonable accuracy. However, several cases exhibit errors ranging from a few millimeters to more substantial deviations. The width estimates show greater scatter, with many cases deviating from the true values by significant margins. The poor estimation of notch width is consistent with the sensitivity analysis, which identified width as the least influential parameter. When a parameter contributes minimally to the response variance, the measured data provide limited information for its inference.

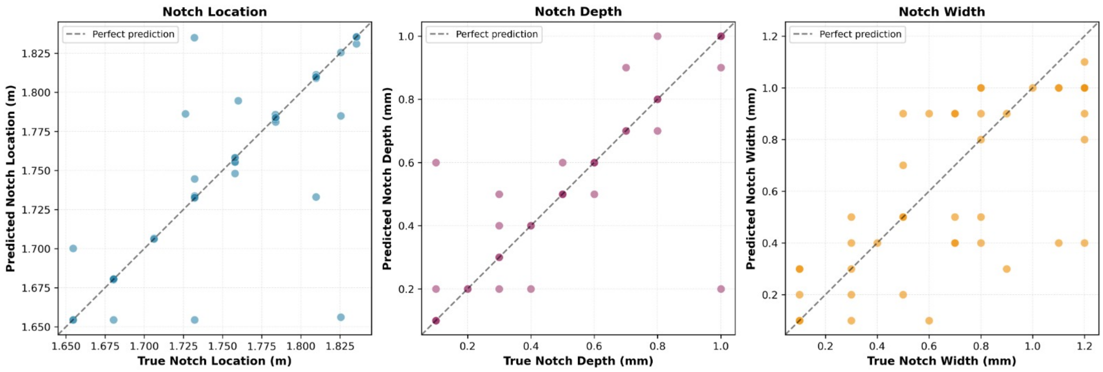


Figure 5.9: MAP estimates versus true parameter values (top row) and corresponding error histograms (bottom row) for notch location, depth, and width across all test cases. Location and depth estimates cluster near the diagonal, while width estimates show larger scatter consistent with low parameter sensitivity.

For location and depth, the estimation errors observed in certain cases can be attributed to two factors. First, reconstruction errors in the forward surrogate introduce systematic biases that propagate to the inverse estimates. Second, the sampling resolution of the training data may be insufficient to capture fine variations in parameter space, leading to interpolation errors for certain parameter combinations. Despite these limitations, the majority of test cases yield location and depth estimates within acceptable tolerances, as can be seen in the normalized error histogram plot Figure 5.10.

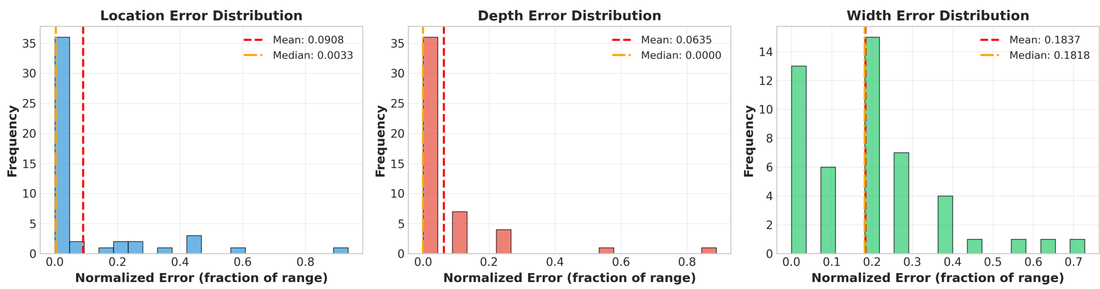


Figure 5.10: Distribution of normalized absolute estimation errors for each notch parameter. Errors are normalized by the respective parameter ranges. Location and depth errors are predominantly below 10%, while width errors span a wider range.

From a practical standpoint, accurate estimation of location and depth is more critical than width estimation. A crack with substantial depth but small width may be visually undetectable yet structurally significant. Conversely, a shallow wide scratch poses less risk to structural integrity. The proposed framework demonstrates capability in estimating the parameters that matter most for structural health assessment.

5.6.3 Posterior distributions: Best-case scenario

Corner plots provide a comprehensive visualization of multi-parameter posterior distributions by combining marginal and joint probability information in a single display. The diagonal panels show the one-dimensional marginal distribution for each parameter, revealing the uncertainty associated with individual parameters when all others are integrated out. The off-diagonal panels display the two-dimensional joint posterior distributions between parameter pairs, which expose correlations and dependencies that single-parameter analyses cannot capture. This visualization becomes particularly valuable in Bayesian inverse problems where parameter interactions can significantly influence the inference results.

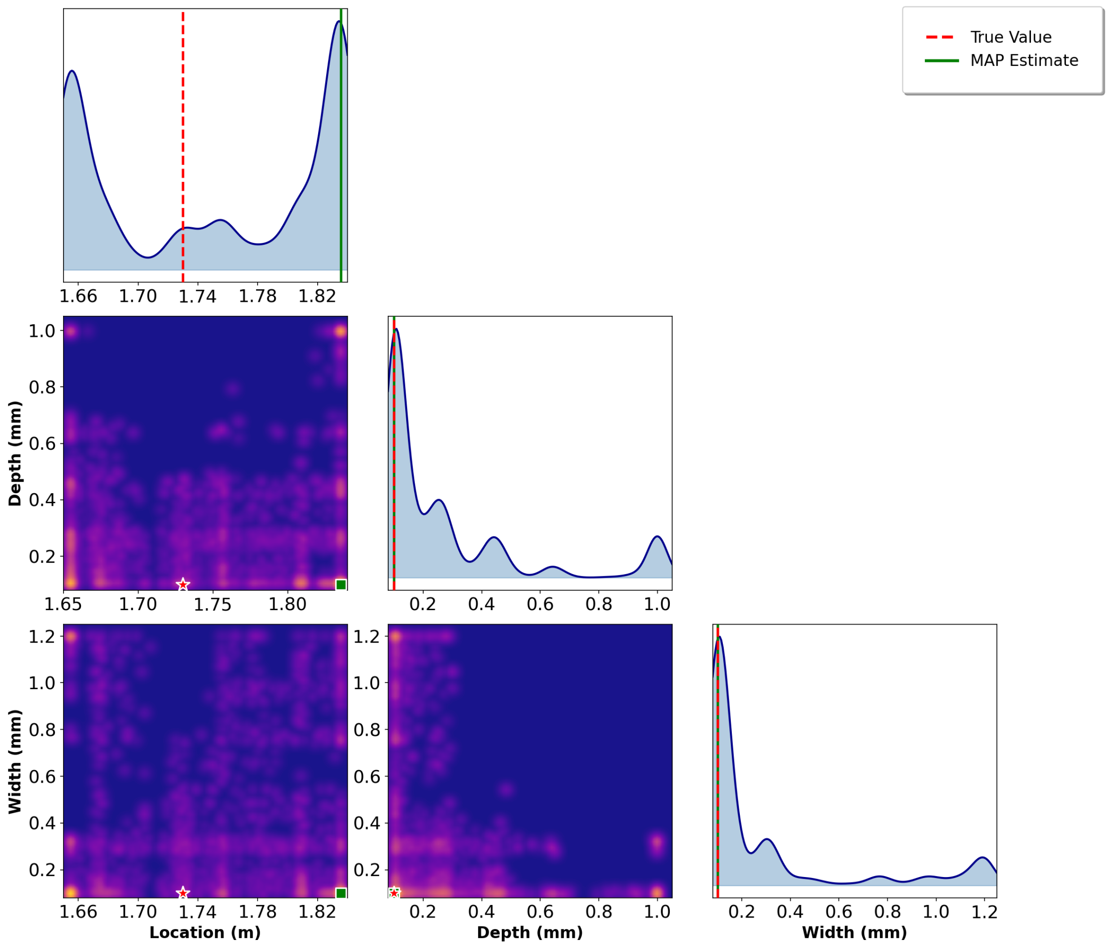


Figure 5.11: Corner plot for a test case with accurate forward prediction ($R^2 = 0.998$). Diagonal panels show marginal posteriors with true values marked by vertical lines. Off-diagonal panels display joint distributions; stars denote true values and squares indicate MAP estimates. The location posterior exhibits multimodality with the true value coinciding with a secondary mode.

Figure 5.11 presents the corner plot for the test case with the best forward prediction accuracy by MFSM on one of the test cases. The depth and width parameters are captured with reasonable accuracy by the MAP estimate. The true values fall within the high-probability regions of the posteriors, indicating that the MCMC sampling has successfully identified the correct parameter space. The location parameter presents an interesting case. The MAP estimate misses the true location, yet the true value coincides with one of the peaks in the posterior distribution. The posterior correctly identifies the true location as a plausible candidate, even though the highest peak mode corresponds to a different value. Such multimodality arises possibly from the errors in the MFSM. Notches at different locations can produce similar scattering patterns under certain conditions, leading the inference algorithm to assign comparable posterior probability to multiple location values that generate nearly equivalent responses.

5.6.4 Posterior distribution: Worst case scenario

Figure 5.12 shows the corner plot for the test case with the worst forward prediction by MFSM. Despite the poor reconstruction quality for this case, the inference algorithm manages to estimate the location and depth with accuracy. The location estimate benefits from information encoded in both the symmetric and antisymmetric wave modes. For depth, the algorithm attempts to match the amplitude characteristics of the scattered portion. Since the surrogate has learned that increasing depth amplifies the scattered wave amplitude, the inference process pushes the depth estimate toward values that produce comparable amplitude levels. When the true depth lies at an extreme of the parameter range, the algorithm cannot exceed this boundary, resulting in an estimate at the parametric limit. This behavior reflects appropriate learning of the monotonic relationship between depth and scattering amplitude.

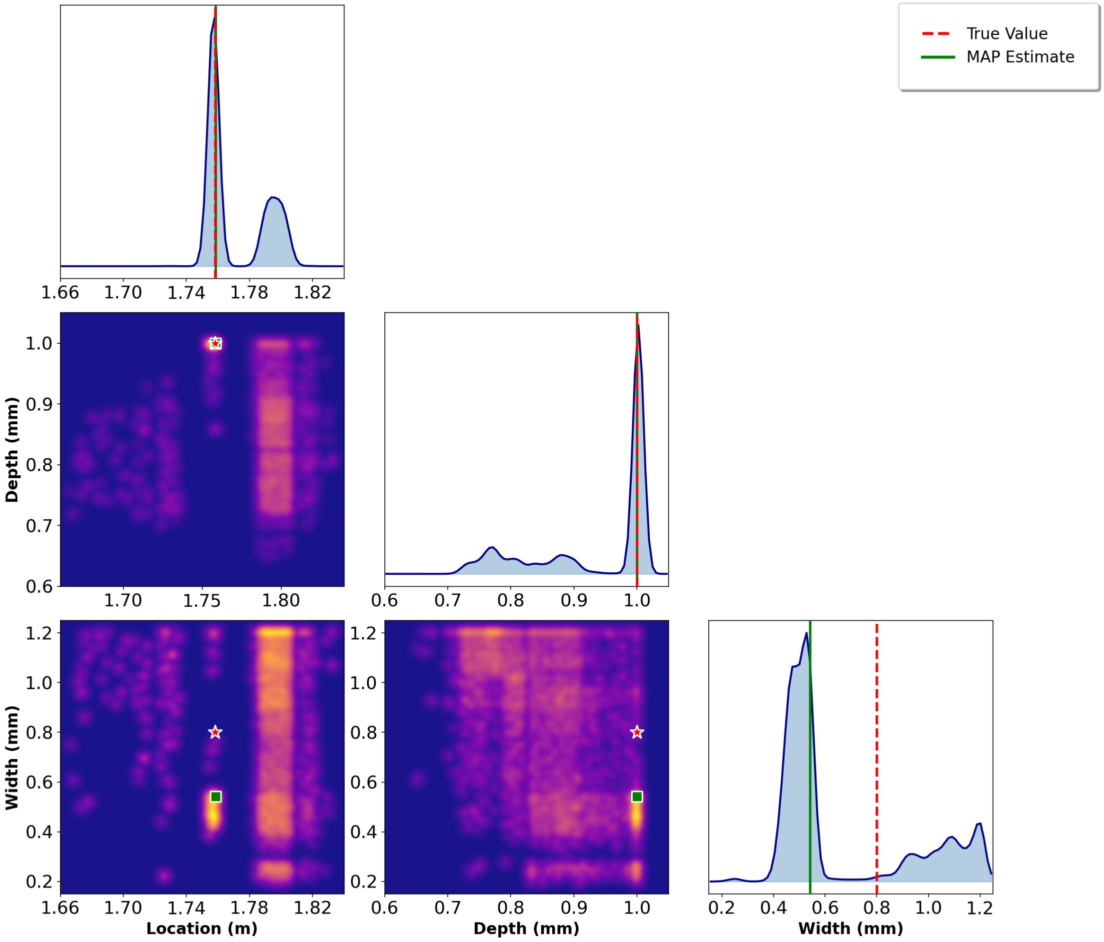


Figure 5.12: Corner plot for the worst forward prediction case ($R^2 = 0.928$). Despite reconstruction challenges, location and depth posteriors concentrate near true values (marked by dashed lines). Width posterior remains diffuse, reflecting low parameter identifiability.

5.6.5 Optimal parameter identification case

Figure 5.13 presents the reconstructed response for the test case with the most accurate parameter estimates.

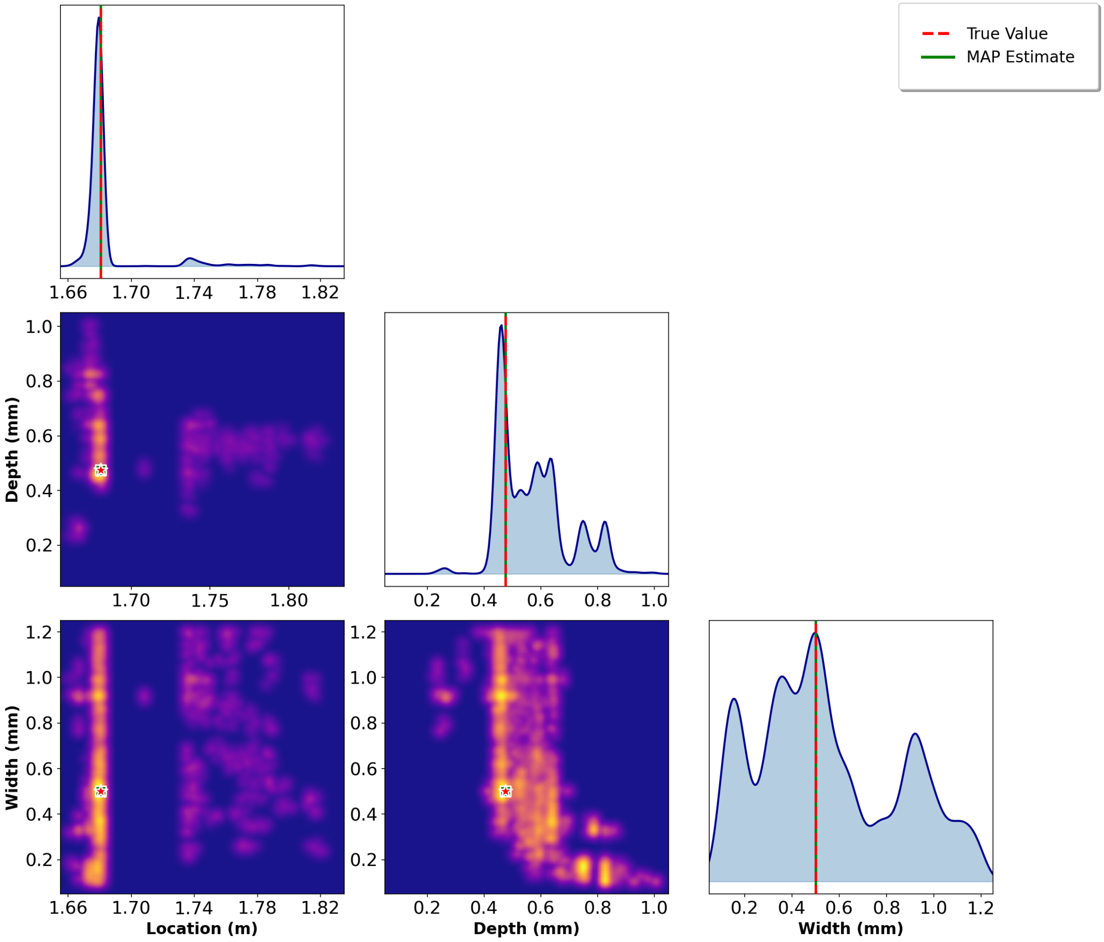


Figure 5.13: Response reconstruction for the test case with the most accurate parameter estimates. The MAP-predicted response (green) closely matches the observed signal (dashed), with narrow uncertainty bounds indicating high surrogate confidence in this parameter region.

The reconstructed response closely matches the target, confirming that accurate parameter estimation translates to accurate response prediction. This case represents a favorable scenario where the surrogate model performs well and the true parameters lie within a well-sampled region of parameter space.

The location parameter shows the highest estimation confidence, with a single sharp peak in the marginal distribution centered near the true value. This narrow posterior indicates strong constraints from the observed wave response. The depth parameter achieves reasonable accuracy despite moderate uncertainty. The marginal distribution contains a dominant mode capturing the true depth, but secondary peaks appear at alternative values. The true depth falls within the highest probability region, indicating successful identification. The width parameter exhibits the broadest posterior distribution with multiple modes of comparable height. Several width values can generate responses matching the observations reasonably well. The true width resides within one of the prominent peaks, but the multimodal posterior reveals an inherent identifiability challenge for this parameter.

5.6.6 Multimodal posterior analysis

Figure 5.14 illustrates a case where the location is estimated at a position nearly opposite to the true value along the beam, yet the estimate corresponds to a legitimate mode in the posterior.

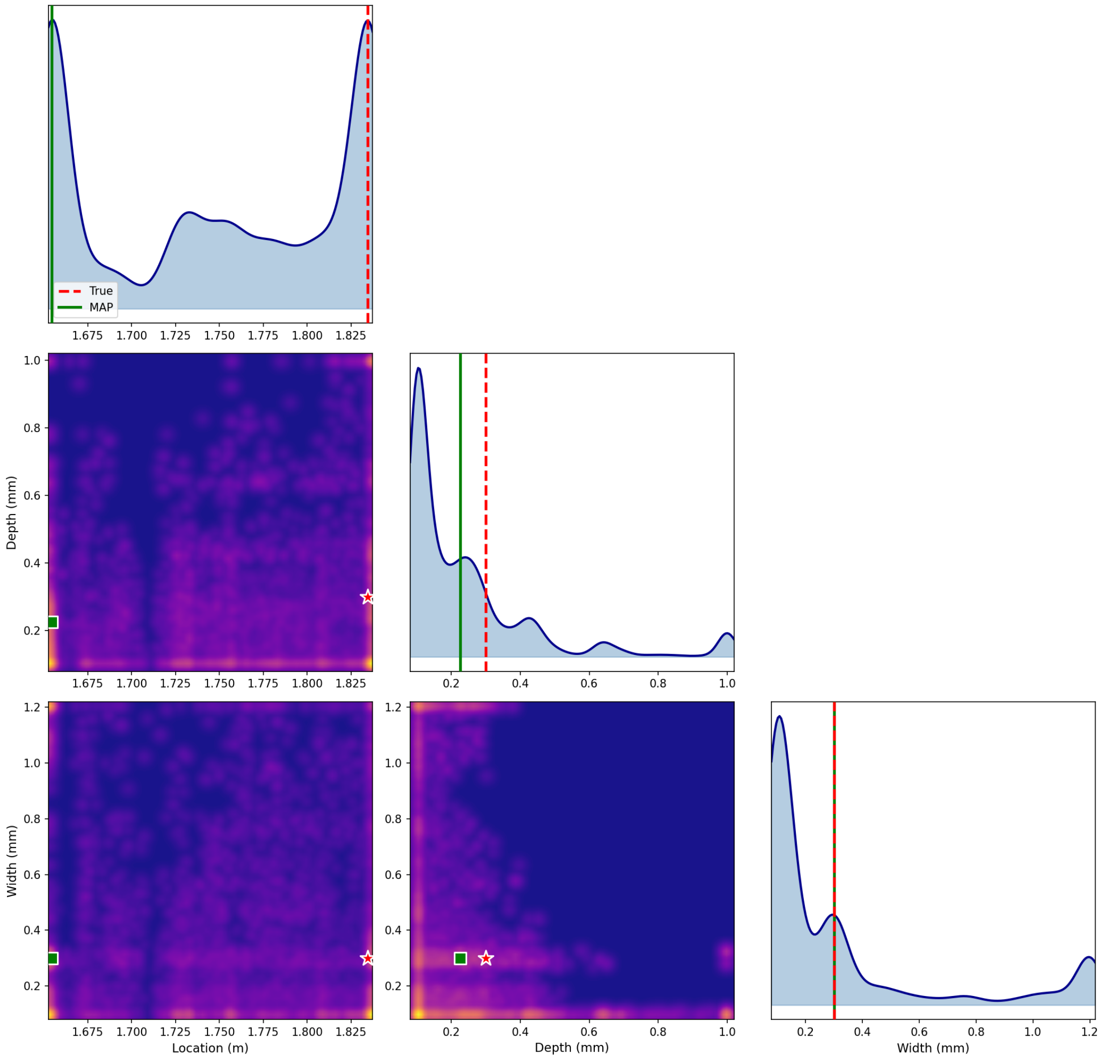


Figure 5.14: A case demonstrating multimodal posterior behavior arising from ill-posedness and approximation error. The MAP location estimate corresponds to an alternative mode producing similar sensor response; the true location appears as a secondary posterior mode, reflecting legitimate ambiguity in the inverse problem.

This behavior exemplifies the inherent ambiguity in wave-based inverse problems. Due to the symmetry of the waveguide and the interaction of multiple propagating modes, notches at different locations can generate similar sensor responses. The Bayesian framework captures this ambiguity through multimodal posteriors. While the MAP estimate may miss the true value, the posterior distribution as a whole contains the correct solution. For practical applications, this suggests that inverse estimates should not be interpreted as definitive point values but rather as distributions that encode both the most likely configurations and the associated uncertainty.

5.6.7 Inverse inference using HFSM

To quantify the value of multi-fidelity learning for inverse problems, the same Bayesian inference framework was applied using the high-fidelity-only surrogate model (HFSM) as the forward model. The HFSM was trained exclusively on 100 high-fidelity samples without low-fidelity pre-training, achieving an R^2 of 0.8703 compared to the MFSM's 0.9693. This lower accuracy reflects the HFSM's failure to adequately learn response reconstruction across the parameter space when trained on limited data.

The inverse problem used identical settings: Differential Evolution initialization with 30 generations, parallel tempering MCMC with 48 chains across 6 temperature levels, and 5000 iterations per chain. Two test cases corresponding to the configurations analyzed for MFSM were selected to enable direct comparison of how surrogate model quality affects posterior distributions.

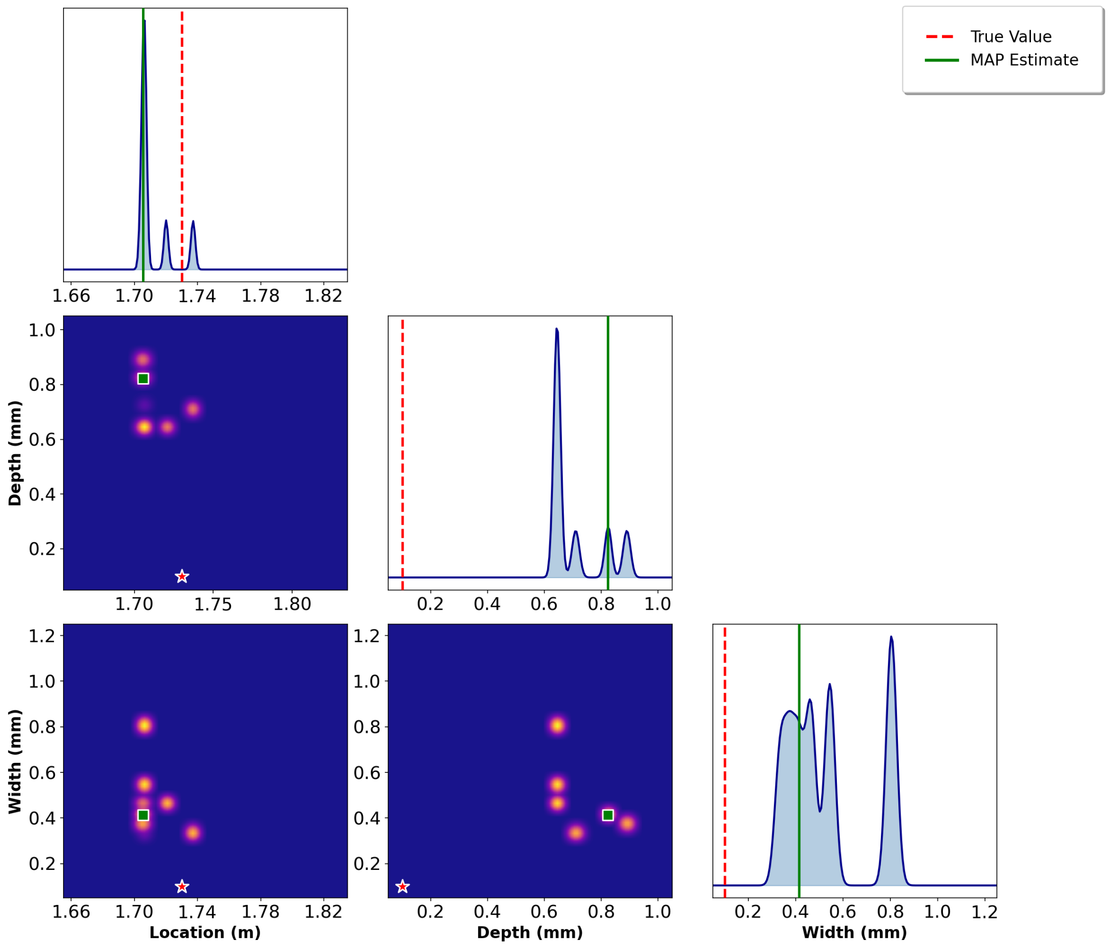


Figure 5.15: Posterior distributions using HFSM for a deep, wide notch case (same configuration as Figure 5.12). Complete parameter identification failure occurs: no posterior modes coincide with true values, illustrating how surrogate inaccuracy propagates to misleading inverse estimates.

Small notch configuration

Figure 5.15 presents posterior distributions for a small notch case matching the configuration from Figure 5.11.

The posteriors fail to capture the true parameters. The location posterior shows sampling in the vicinity of the true value, though no posterior mode peaks at the true location. The MAP estimate lies elsewhere. For depth and width, the posteriors concentrate in regions distant from the true values. The inference algorithm fails because the HFSM produces inaccurate response predictions across the parameter space. When the surrogate misrepresents the forward model, the likelihood function from Equation 4.4 assigns high probability to incorrect parameter values that happen to match the observed response through compensating errors.

Deep and wide notch configuration

Figure 5.16 shows posteriors for a deep, wide notch case corresponding to Figure 5.12.

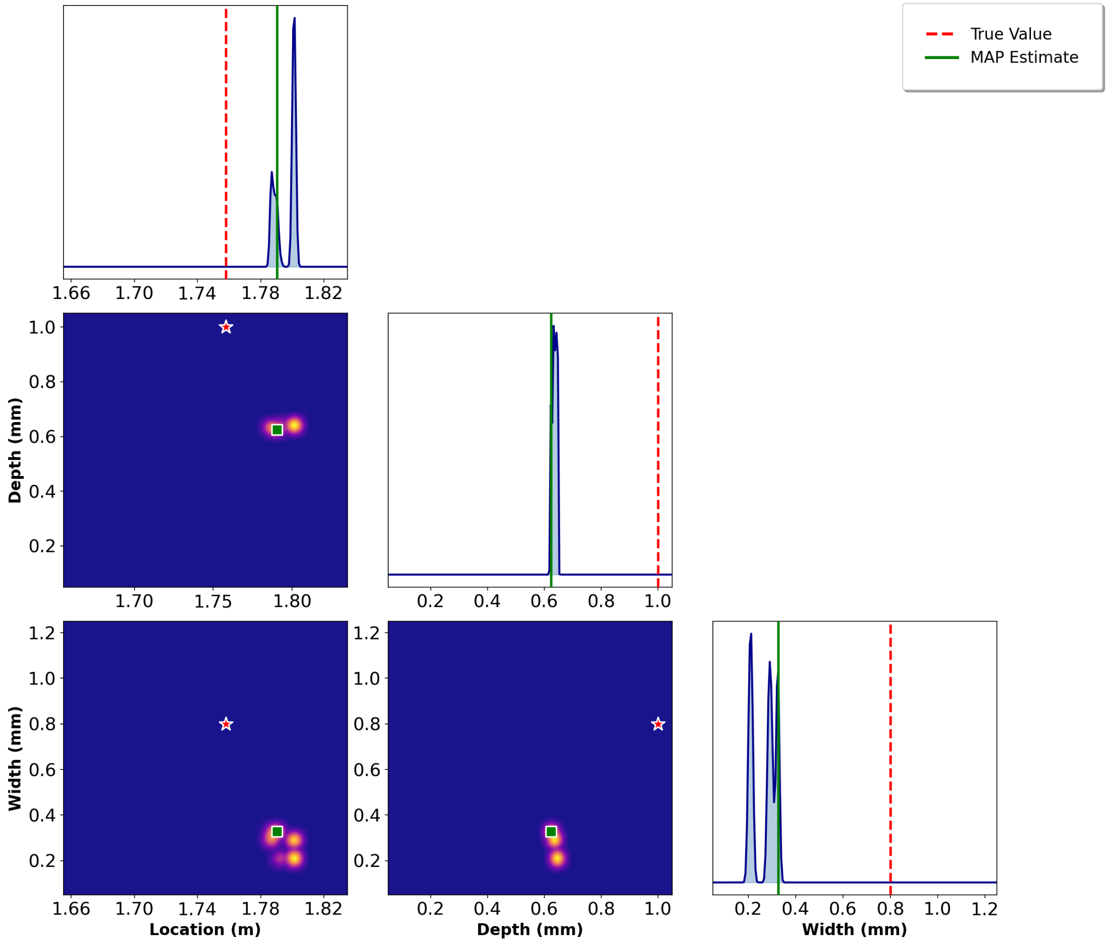


Figure 5.16: Posterior distributions using HFSM for deep and wide notch case. The posteriors fail to identify true parameters, with posterior mass concentrated away from true values.

This case shows complete failure of parameter identification. None of the three parameters are captured by the MAP estimate or by any significant posterior mode. The location posterior exhibits multimodal structure with peaks scattered across the domain, none coinciding with the true location. The depth posterior concentrates at values considerably different from the true depth. The width posterior is also far away from the true value. These failures stem directly from the HFSM's poor reconstruction quality, particularly for parameter combinations near the boundaries of the training domain where the surrogate has insufficient data coverage.

Comparison with multi-fidelity surrogate and reason for failure

The contrast with MFSM results reveals why multi-fidelity learning is necessary. In Figure 5.11, the MFSM posterior captures the true location as a distinct mode and gives accurate depth estimates despite multimodality. In Figure 5.12, the MFSM estimates location and depth with high accuracy even for the challenging deep notch case. The HFSM fails in both cases.

The root cause is inadequate response reconstruction. The HFSM, trained on only 100 samples, cannot learn the complex parameter-to-response mapping across the full parameter space. Without the initial feature learning phase provided by 750 low-fidelity samples, the autoencoder struggles to develop robust latent representations. The HFSM produces systematic reconstruction errors that bias the likelihood function, assigning spuriously high probabilities to incorrect parameters through error cancellation rather than physical accuracy. The pointwise variances estimated from only 100 training samples provide unreliable uncertainty quantification that fails to adequately inflate likelihood variance in poorly-learned regions.

These results confirm that surrogate quality is necessary for inverse problem success. A surrogate with $R^2 = 0.87$ proves insufficient while $R^2 = 0.97$ enables parameter identification. Multi-fidelity learning provides value beyond simply expanding the training set. The low-fidelity data encode physical constraints that guide the surrogate toward learning generalizable wave propagation features, enabling accurate inference even with limited high-fidelity samples.

Chapter 6

Conclusions and Future Work

This thesis develops an *uncertainty-aware multi-fidelity surrogate framework* for inverse damage characterization in guided-wave-based structural health monitoring. The work demonstrates that transfer learning between low-fidelity zigzag beam theory and high-fidelity finite element simulations enables practical Bayesian inference when computational resources are limited.

6.1 Summary of Findings and Contributions

Transfer learning proves necessary for surrogate-based inverse inference under data scarcity. A surrogate trained exclusively on 100 high-fidelity samples achieves $R^2 = 0.87$ and fails to support parameter estimation. The same architecture trained first on 750 low-fidelity samples then fine-tuned on identical high-fidelity data reaches $R^2 = 0.97$ and enables successful inference. This 10 percentage point improvement translates directly to posterior quality. The high-fidelity-only surrogate produces posteriors that miss true parameters in both test cases examined, while the multi-fidelity surrogate captures true values within the posterior distribution.

The framework treats surrogate approximation error as a distinct uncertainty source alongside measurement noise. Gaussian latent-space modeling with covariance estimated from residuals quantifies surrogate-induced uncertainty. Monte Carlo sampling propagates this uncertainty to response space, preventing artificially confident posteriors. The resulting likelihood function incorporates both measurement noise and surrogate approximation error, providing rigorous uncertainty quantification.

Differential Evolution initialization substantially improves MCMC convergence. Running 30 generations of Differential Evolution and seeding some chains from the resulting population provides smart initialization while preserving uninformed prior assumptions. Markov chains initialized near Differential Evolution solutions explore local structure while prior-initialized chains prevent premature concentration on spurious modes.

Notch location and depth are identifiable from guided wave responses, while width estimation remains unreliable. Sobol sensitivity analysis confirms location and depth dominate the response. Deep notches produce strong scattering regardless of width, making width variations difficult to detect. Location and depth posteriors concentrate around true values in most test cases. Width posteriors remain diffuse, correctly reflecting low sensitivity.

Multimodal posterior distributions observed in the Bayesian inversion arise from a combination of physical non-uniqueness in guided-wave propagation and approximation errors introduced by the multi-fidelity surrogate model. From a physical standpoint, wave propagation in waveguides can lead to non-unique inverse mappings, where some damage configurations produce similar dispersive responses at the sensor due to mode conversion, interference, and limited spatial sensing. Such ambiguity is inherent to the physics of the problem and cannot be eliminated by improved numerical resolution alone. At the same time, the accuracy of the multi-fidelity surrogate model plays a critical role in shaping the posterior landscape. Residual surrogate approximation errors—particularly under limited high-fidelity training data—can blur distinctions between competing parameter configurations, amplifying or introducing secondary posterior modes. In this sense, some degree of multimodality reflects the current limitations of the surrogate rather than fundamental physical equivalence. The Bayesian framework provides a principled mechanism to represent both sources of uncertainty. Rather than relying solely on a single maximum a posteriori estimate, the full posterior distribution reveals multiple plausible damage scenarios consistent with the observed response and the surrogate uncertainty model. This probabilistic representation is essential for informed decision-making, as it explicitly exposes ambiguity and confidence levels, while also highlighting directions for surrogate improvement through targeted high-fidelity data enrichment.

The computational speedup makes Bayesian inference tractable. High-fidelity simulation requires 38 minutes while surrogate prediction takes 50 milliseconds, yielding a speedup exceeding 400000. The DE-PT-MCMC framework evaluates the forward model approximately 210000 times, requiring 30 minutes with the surrogate versus over several days with direct simulation.

6.2 Current Limitations and Assumptions

The framework assumes constant environmental conditions. Temperature variations and operational loads are excluded, though both affect wave propagation in practice. The diagonal covariance approximation may underestimate temporal correlations between response components. More sophisticated uncertainty models could improve calibration at increased computational cost.

The methodology addresses only rectangular notches in homogeneous beams. Real damage exhibits complex shapes including irregular fatigue cracks, delaminations, and distributed corrosion. Extension to these damage types requires reformulating the parameterization and potentially incorporating additional measurement data.

Multiple simultaneous damage sites introduce combinatorial complexity. The three-parameter vector describing one notch grows linearly with damage count, yielding high-dimensional posteriors requiring enhanced sampling strategies.

Experimental validation is absent. All results derive from numerical simulations with synthesized measurements. Real experimental data contains modeling errors, sensor coupling variations, and boundary imperfections not captured in simulations. Laboratory experiments on beam specimens with controlled notches would provide necessary validation before field deployment.

6.3 Recommendations for Future Research

Experimental validation using physical beam specimens represents the immediate priority. Aluminum beams with realistic notches and bonded piezoelectric transducers would provide controlled test cases. Comparing experimental measurements to surrogate predictions would quantify modeling error and validate the uncertainty quantification strategy.

Heteroscedastic uncertainty models could capture parameter-dependent surrogate accuracy. The current framework estimates a single covariance matrix from all validation residuals. A heteroscedastic model would estimate input-dependent covariance using Gaussian processes or neural network techniques, preventing overconfident posteriors in poorly-learned parameter regions.

Extension to composite materials introduces anisotropy and layered structures. Zigzag theory provides appropriate low-fidelity models for composite laminates. However, parameter dimensionality increases substantially when accounting for layer-specific degradation, requiring enhanced sampling strategies.

Adaptive sampling could optimize expensive high-fidelity data collection. Active learning would identify which parameter combinations to simulate based on current surrogate uncertainty. Acquisition functions balancing exploration and exploitation would guide sequential sample selection, potentially reducing required simulations.

Multiple damage scenarios could be addressed through hierarchical Bayesian models with trans-dimensional MCMC. The prior would include discrete variables representing damage site count, enabling joint inference of how many sites exist and their characteristics.

Integration with structural reliability assessment would provide decision-relevant outputs. Propagating parameter uncertainty through capacity models would yield probabilistic estimates of remaining strength and failure probability, justifying monitoring costs through value of information calculations.

The methodology generalizes beyond guided wave monitoring. Any inverse problem with available fidelity hierarchies could benefit from this approach, including vibration-based damage detection, thermal imaging for subsurface defects, and electromagnetic material characterization. The key requirement is a fast low-fidelity model correlating sufficiently with high-fidelity physics to support effective transfer learning.

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